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Joseph Henry Briggs

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# Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Chapter ??: Not started. Will write last/almost last
- Chapter ??: Apparently this is a good chapter...
- Chapter ??: Passed first drafting. Very Minor corrections to be made.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
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- Chapter ??: Passed first drafting. Still to apply Ken and bryan comments
- Chapter ??: Not written, vaguely planned
- Chapter 1: Significant amounts of thought has already gone into this.
- Chapter 1.11: Not started. Will write last/almost last

# Chapter 1

## Characterisation of InGaAs

### Photodiodes for detecting 2 $\mu$ m Light

#### 1.1 Introduction

Future gravitational wave detectors may operate with 2  $\mu$ m wavelength light. This means that new readout photodiodes could be needed. Extended InGaAs photodiodes could be used for this. In this chapter, the working principles and properties of extended InGaAs is discussed in relation to the requirements of a third generation gravitational wave detector. The dark noise and quantum efficiency of off-the-shelf photodiodes was investigated with the aim of understanding general trends in the behaviour of extended InGaAs photodiodes. It was found that to get a high quantum efficiency, the photodiodes must be operated with a bias. However, if the photodiodes were used with a high bias, their dark noise could exceed the shot noise level of the detected light. The dark noise could be reduced significantly by cooling the photodiodes and so future detectors may feature cooled readout photodiodes.

However, at the time of writing (2020) and to the authors knowledge, commercially available InGaAs photodiodes for detecting  $2\text{ }\mu\text{m}$  light that are suitable for use as the readout photodiode in a gravitational wave detector do not exist.

The current generation of gravitational wave detectors will be limited by the thermal noise due to the Brownian motion of the molecules that make up the mirror coatings. Since these thermally driven fluctuations are a limiting source of noise [1], proposed future gravitational wave detectors, such as the Einstein Telescope [2] and LIGO Cosmic Explorer [3], will use cryogenically cooled mirrors.

However, cryogenically cooled mirror coatings will not be compatible with the current laser wavelength, which is  $1\text{ }\mu\text{m}$ , and so future gravitational wave detectors may use either a  $1.5\text{ }\mu\text{m}$  or  $2\text{ }\mu\text{m}$  wavelength laser.  $2\text{ }\mu\text{m}$  is of interest as there have been promising designs for low noise mirror coatings that are optimised for this wavelength.

As InGaAs is a widely used semiconductor for detecting  $1\text{-}1.6\text{ }\mu\text{m}$  light, suitable InGaAs photodiodes were available for detecting the signal-carrying light in the current generation of gravitational wave detectors. These photodiodes are low noise, linear, can operate at room temperature, can detect tens of milliwatts of light, and have  $> 99\%$  quantum efficiency.

As the current InGaAs photodiodes are not sensitive to  $2\text{ }\mu\text{m}$  light, ‘extended’ InGaAs photodiodes may be used instead. These photodiodes are extended in the sense that their cut-off wavelengths are above  $1.6\text{ }\mu\text{m}$ . To achieve this longer cut-off wavelength, extended InGaAs contains a higher percentage of indium compared to

conventional InGaAs photodiodes; however, this extra indium comes at the expense of the structural integrity of the lattice. This can cause the quality of the photodiode to be degraded. Previous experiments have been limited by the quantum efficiency of the extended InGaAs photodiode that was used [4, 5]. To obtain low enough electronic noise in these experiments, the authors of [4] used the photodiodes without a bias – this came at the expense of the photodiodes quantum efficiency.

Alternative photodiode materials that could be used include HgCdTe and InSb. HgCdTe photodiodes with quantum efficiencies  $> 99\%$  are in the design stages of being made, however there are still manufacturing challenges to overcome. InSb detectors have high quantum efficiency [6], however this material would only be suitable for wavelengths above  $2.6\text{ }\mu\text{m}$ .

## 1.2 What is an Extended InGaAs Photodiode

In this section, a qualitative explanation of how photodiodes work will be given. A description of what an extended InGaAs photodiode is and the lattice-mismatch related defects associated with them will be given. The relationship between defects in InGaAs photodiode and properties such as dark current, dark noise and quantum efficiency of photodiodes will be discussed.

A photodiode is a square law detector that produces a current that is proportional to the power of the light it is detecting. As photodiodes are low-passed square law devices, they effectively demodulate the light field and produce a signal with a DC component and components proportional to the frequency and amplitude modulations of the field [?]. Usually, photodiodes are of the order 1 mm in diameter,

have a relatively high quantum efficiency and a fast response. They are two-terminal devices with an anode and a cathode. Photodiode often follow a p-doped, intrinsic, n-doped (PIN) semiconductor structure. PIN photodiodes can detect up to tens of milliwatts of light power.

The basic physics of a PIN photodiode can be qualitatively understood by considering a simpler pn junction. A pn junction consists of two types of semiconductor: p-doped and n-doped. In the p-doped region, holes (the majority charge carriers) are free to move and there are negatively charged ions (acceptors) that are fixed in position. In the n-doped region, electrons (the majority charge carriers) are free to move and there are positively charged ions (donors) fixed in position. In equilibrium, the number of holes is equal to the number of electrons. The free electrons/holes will diffuse across the interface between the p-doped and n-doped region. This causes the p-doped and n-doped regions to become charged. This results in a region of the pn junction where there are no free electrons and holes. The region of the device containing no free carriers is called the depletion region.

When a photon is absorbed by a pn junction, a free electron-hole will be generated. If this photon is absorbed within the depletion region of the photodiode, the electron and hole will be separated due to the electric field and will generate a current in the circuit connected to the photodiode. However, if a photon is absorbed outside the depletion region, then the electron-hole pair will recombine and will not contribute to the current in the circuit. The quantum efficiency of a photodiode is the proportion of photons that are converted into photocurrent. The relationship between quantum

efficiency,  $\eta$ , light power,  $P_{\text{EM}}$ , and photocurrent,  $I_p$  is

$$I_p = \eta \frac{q\lambda_{\text{EM}}}{hc} P_{\text{EM}}, \quad (1.1)$$

where  $\lambda_{\text{EM}}$  is the wavelength of the light,  $h$  is Planck's constant and  $c$  is the speed of light.

To widen the depletion region, an external voltage can be used. When a *reverse bias* – a positive voltage across the cathode and anode – is applied to the photodiode, there is a larger electric field difference at the interface between the p-doped and n-doped region. This increases the volume of the photodiode in which absorbed photons will generate photocurrent.

To further increase the width of the depletion region, a layer of intrinsic semiconductor can be used to separate the p- and n-doped region. An intrinsic semiconductor has no doping and the number of electrons equals the number of holes within it. An electric field will be present in a larger volume of material and so more photons will be converted into electrons which leave the photodiode.

InGaAs is a widely used semiconductor for making photodiodes. Typically, the indium content is around 53%, sometimes this is made clear by writing  $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ , so that the InGaAs can be lattice matched to the InP substrate<sup>1</sup> on which the photodiode is grown on. With this amount of indium, the InGaAs photodiode will suffer from few lattice mismatch defects and so the photodiodes created with it will be of high quality. Typically they are sensitive up to wavelengths around 1.6  $\mu\text{m}$ .

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<sup>1</sup>Ge can also be used

Longer wavelength sensitivity can be achieved by increasing the photodiode's indium content [7]. These photodiodes are often called 'extended' and can be sensitive for wavelengths up to 3  $\mu\text{m}$ . One of the main issues faced when creating an extended InGaAs photodiode is the lattice mismatch between the InGaAs and the InP substrate on which it is grown [8] as lattice mismatches introduce defects.

Defects caused by the strain between the InGaAs and InP layers of the photodiode generate excess noise, dark current and reduce its quantum efficiency [9]. The photodiode can be made thinner to reduce this strain, although this reduces the quantum efficiency. The critical thickness, that is the thickness above which strain degrades the crystal, for extended InGaAs is of the order 0.1 nm [8] whereas the absorption length (of the order of  $10^4\text{cm}^{-1}$  [10]) requires the photodiode to be several microns thick for it to have the high quantum efficiency that is required for the detection of gravitational waves (see Section 1.3). Additionally, because InGaAs is not a good conductor of heat, a thicker amount of InGaAs can lead to lower power threshold before the photodiode is thermally destroyed [11].

These defects manifest as abrupt changes in the crystal structure known as misfit dislocations (e.g. [12] [13]). Defects produce trap states and these give rise to a generation-recombination dark current [14], thus defects are the source of  $1/f$  noise [15]. To obtain high quantum efficiency the defect level must be low [16]. Defects increase the doping level of the absorbing layer of the photodiode which causes the depletion region to become smaller, thus causing a decrease in quantum efficiency. For indium doping levels other than 53%, buffer layers of InGaAs between the absorbing layer and the substrate can be used to mitigate the lattice

mismatch [14].

InGaAs that is optimised for 2  $\mu\text{m}$  has a refractive index,  $n$ , of approximately 3.6 [10]. Assuming a single quarter-wave layer was to be applied, the film would need a refractive index that was  $n_f = \sqrt{3.6}$  and be 600 nm thick. The idea of using this anti reflective coating as a passivation layer is presented in [17].

The dark noise and dark current of a photodiode has several contributors. In low bias regimes, the dark current is driven by the diffusion of minority carriers [16, 18]. However, the thermally driven generation-recombination of electron/hole pairs drives the dark noise at low biases [15]. Therefore, the dark current does not predict the dark noise, however as both of them increase with higher levels of defects, a photodiode with a high dark current will have a high dark noise. When the bias is large enough, electrons can tunnel across the junction leading to a great increase in dark current.

The noise of the current made by the photodiode can be modelled as

$$i_{\text{rms}} = \sqrt{2qi + \frac{4k_B T}{R_{\text{PD}}} + \left(\frac{i_c}{f^\beta}\right)^2}. \quad (1.2)$$

The first term under the square-root,  $2qi$ , is the shot noise associated with the DC current  $i$ . The second term,  $\frac{4k_B T}{R_{\text{PD}}}$ , is the Johnson-Nyquist noise due to the shunt resistance of the photodiode,  $R_{\text{PD}}$ ; this term can become significant when the shunt resistance is low, as it is in some extended InGaAs photodiodes, and microwatt levels of light is being detected. The final term is Hooge's law of noise in semiconductors [15];  $i_c$  is the characteristic noise level and  $\beta$  is usually  $0.5 \pm 0.1$ .

## 1.3 Requirements for the Readout Photodiodes in a Gravitational Wave detector

The following section will motivate the quantum efficiency, power handling and dark noise requirements for a photodiode that would be used as the readout photodiode for the third generation of gravitational wave detectors.

The photodiodes used in the readout of a gravitational wave interferometer should have a quantum efficiency  $> 99\%$ . Losses within the interferometer decrease its SNR as they diminish the level to which the signal light is squeezed and so high quantum efficiency is required to detect the baseline level of  $\sim 10$  dB of squeezed light that future gravitational wave detectors will use. Usually, if a bigger signal is required, the gain can be increased by using a larger resistor in the feedback path of the transimpedance amplifier or by using an avalanche photodiode, however neither of these methods would work in a gravitational wave detector as both would boost the noise and the signal in proportion with each other.

Current gravitational wave detectors operate in the 1-10 mW range at the output [22] to ensure that the junk light from other modes is a small portion of the total signal. As  $q\lambda_{\text{EM}}/hc$  is of the order unity, the photocurrent will be assumed to be  $\sim 10$  mA and so any new photodiode should be able to produce  $\sim 10$  mA of photocurrent without saturating. If heterodyne readout was to be used, then the DC light power could be reduced.

Photodiodes for gravitational wave detectors should have low  $1/f$  dark noise so that the interferometers are not limited in their detection bands by electronic noise.

Third generation ground based gravitational wave detectors will be quantum noise limited down to about 10 Hz [19], so the  $1/f$  noise must be below the shot noise of the photocurrent above this frequency. For a 10 mA photocurrent, the shot noise is  $6 \times 10^{-11} \text{ A}_{\text{rms}}/\sqrt{\text{Hz}}$ . Below 10 Hz other rapidly rising sources of noise will limit the sensitivity of the Einstein Telescope [2] and Cosmic Explorer [23]

The  $1/f$  noise could be avoided by using heterodyne detection, however a conventional heterodyne detection scheme would increase the shot noise level of the interferometer. Heterodyne detection can decrease the SNR of the interferometer if the mixing of the modulation and demodulation waveforms was not time independent [20]. Although an optimal choice of modulation-demodulation waveform can recover the SNR by increasing the signal, the benefits of using squeezed light would be lost because the shot noise would be enhanced, compared to homodyne readout, by vacuum noise at frequencies that contain no signal entering into the measurement. However, it has been proposed that the increase in noise associated with heterodyne readout could be avoided by the use of a broadband squeezer [21].

## 1.4 Selected Off-the-shelf Extended InGaAs Photodiodes

The quantum efficiency, saturation limit, dark current and dark noise was characterised in a selection of commercially available photodiodes. The photodiodes that were tested and their properties according to their datasheets<sup>2</sup> are listed in table

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<sup>2</sup>Datasheets often quote the noise equivalent power (NEP) of a photodiode. The NEP is the noise spectral density of the dark current generated by the detector divided by the responsivity of the photodiode. For the Hamamatsu g12183 series photodiodes, the responsivity is  $\sim 1 \text{ A/W}$

1.1. According to each of the photodiodes datasheets, none of them have a quantum efficiency of  $> 99\%$ ; these photodiodes have quantum efficiencies around  $80\%$ . However, the quantum efficiency of a photodiode can be improved by increasing the bias voltage (see Section ??). Increasing the bias increases the dark current and dark noise of the photodiode and so for each photodiode a maximum reverse bias voltage was found. The benchmark used was the photodiode's dark noise at  $10\text{ Hz}$ ; the maximum useful bias was determined by the voltage where the dark noise was equal to the shot noise of a  $10\text{ mA}$  current. The bias voltage also affects the saturation limit of the photodiode. If a photodiode was found to have a low saturation threshold ( $\sim 1\text{ mA}$ ) then it would not be fit for use in a gravitational wave detector.

## 1.5 The effect of Reverse Bias on the Quantum Efficiency of an Extended InGaAs Photodiode

The relationship between the reverse bias and quantum efficiency of an extended InGaAs photodiode, the Thorlabs FD10D, was investigated. The FD10D was exposed to a variety of beams so that saturation effects would be visible. Two styles of measurements were used: one where the reverse bias and the response of the photodiode were measured at DC, and a lock-in style measurement where the quantity

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and the NEP is  $\sim 10^{-12}\text{ W}/\sqrt{\text{Hz}}$ , so the noise, as claimed by the datasheet, is  $\sim 10^{-12}\text{ A}/\sqrt{\text{Hz}}$ . This would seem sufficiently low to be shot noise limited down to  $\sim 1\mu\text{W}$ . However, this number does not capture the behaviour of the photodiode at illumination levels of several milliwatts where a bias is required to ensure the photodiode does not saturate. As the noise of these photodiodes is not flat (see Appendix A), NEP should be specified as a function of frequency. The noise and responsivity are both functions of reverse bias and so the NEP is a function of bias. Another quantity related to the NEP is the specific detectivity; this incorporates the area and bandwidth of the photodiode with the NEP.

| Photodiode  | Peak quantum efficiency (%) | Cut-off wavelength <sup>a</sup> (μm) | Diameter (mm) |
|-------------|-----------------------------|--------------------------------------|---------------|
| FD10D       | 80 <sup>b</sup>             | 2.5                                  | 1             |
| G12183-010k | 77 <sup>b</sup>             | 2.5                                  | 1             |
| G12183-005k | 77 <sup>b</sup>             | 2.5                                  | 0.5           |
| IG26x500S4i | 86 <sup>c</sup>             | 2.5                                  | 0.5           |
| IG24x500S4i | 84 <sup>c</sup>             | 2.4                                  | 0.5           |
| G12182-005k | 84 <sup>d</sup>             | 2.1                                  | 0.5           |
| G12182-010k | 84 <sup>d</sup>             | 2.1                                  | 1.0           |

<sup>a</sup>  $\lambda_{EM}$  for  $\eta = 50\%$    <sup>b</sup>  $\lambda_{EM} = 1.9\mu\text{m}$    <sup>c</sup>  $\lambda_{EM} = 2.0\mu\text{m}$    <sup>d</sup>  $\lambda_{EM} = 1.6\mu\text{m}$

Table 1.1: Datasheet information for the photodiodes that were tested. Note that none of the photodiodes has a quantum efficiency  $> 99\%$ , however measurements of dark noise and dark current and investigations into their optimal operating conditions is still of interest. This selection of photodiodes covers a range of manufacturers, sizes and cut-off wavelengths.

$\frac{\Delta\eta}{\Delta V_B}$  was measured. The quantum efficiency should be independent of frequency if the photodiode is operating in a time-independent, linear fashion and so the two styles of measurements should yield the same outcome. To prove that the increase in signal was a result of the photodiode's quantum efficiency increasing, and not some other mechanism e.g. an avalanche effect, the photodiode was biased such that the dark noise was sufficiently low (see Section ?? for details.) and the noise of the photocurrent was compared to the level expected for shot noise.

### 1.5.1 Method

To measure the photocurrent produced by the photodiode, a transimpedance amplifier which could produce shot noise limited signals for 10 mA photocurrents was used. This circuit is shown in Appendix B.2. The circuit allowed for the reverse bias

of the photodiode to be set externally via a low-passed bench-top power supply.

A 1550 nm wavelength laser (Pure Photonics PPCL300) was used to illuminate the FD10D. The light from the laser was split by a 50:50 beam splitter so that the laser could be amplitude stabilised. The in-loop photodiode was a high performance InGaAs photodiode (C3061GH). The FD10D was the out-of-loop photodiode. The laser was amplitude stabilised by actuating on the laser photodiode's current. Shot noise limited measurements could be made above 1 kHz. A half waveplate and polarising beam splitter were used to control the power level of the light incident on the FD10D, and a lens was used to set the size of the beam.

The FD10D was exposed to a variety of beams to investigate the possibility that current density related phenomena could affect the results. This was done by changing the size and power of the beam. As the FD10D has a diameter of 1 mm, the laser beam should not have a width larger than 330  $\mu\text{m}$  for there to be  $< 1\%$  losses due to clipping. As gravitational wave detectors operate with 1-10 mW of power, the FD10D was tested with powers in this range. The methods in this section can only be used to measure relative increases in quantum efficiency; measurements of absolute quantum efficiency require [find ref ??](#) additional apparatus.

To measure the effect that changing the reverse bias voltage has on the quantum efficiency of a photodiode, two approaches were used. It is straight forward to determine the change in quantum efficiency by measuring the change in photocurrent at DC. Alternatively, one can perform a lock-in style measurement. If the quantum efficiency depends on the reverse bias of the photodiode, then modulating the reverse bias,  $V_B$ , at a frequency  $f$  will result in  $\eta$  modulating at  $f$  and so the measurement

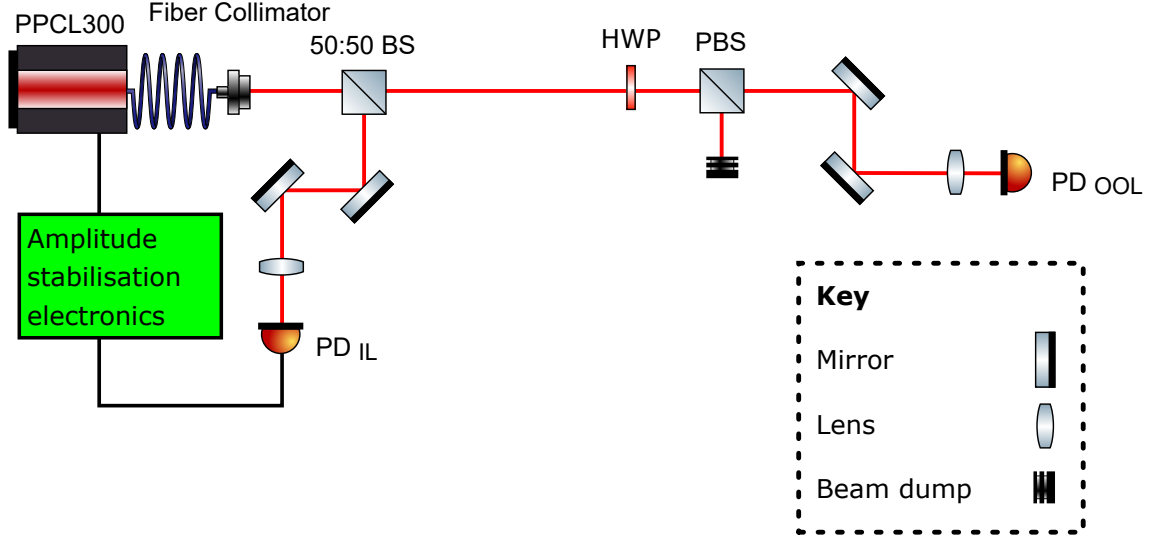


Figure 1.1: Optical layout used for measuring the effect that increasing the reverse bias of a photodiode has on its quantum efficiency and saturation limit. The out-of-loop photodiode, PD<sub>OOL</sub>, was the photodiode under test. A 1550 nm laser (PPCL300) was used to test the optical properties of the photodiode. A combination of half wave plate (HWP) and polarised beam splitter (PBS) allowed for the power of the beam to be set and a lens was used to set the size of the beam on the photodiode under test. Amplitude stabilised light was needed to ensure the increase in photocurrent seen when the bias increased was due to an increase in quantum efficiency. The photodiode circuit (see Appendix B.2) allowed for shot noise limited measurements of currents around 10 mA and for the bias to be selected.

will be of  $\Delta\eta/\Delta V_B$ . The input modulation  $\Delta V_{\text{in}} = \Delta V_B$  and the output modulation is  $\Delta V_{\text{out}} \cos \theta = \Delta\eta V_{\text{TIA}}(V_B)$ , where  $V_{\text{TIA}}(V_B)$  is the DC signal from the photodiode and  $\theta$  is the phase between the input and output modulation.

As 0.6 V reverse bias was the highest bias that gave an acceptable level of dark noise (see Figure 1.5), the noise of the photocurrent was measured at a region where the noise of the laser was limited by shot noise. The expected level of noise,  $n$ , in units of  $A_{\text{rms}}/\sqrt{\text{Hz}}$ , is

$$n = \sqrt{2qI_{\text{OOL}} + 2qI_{\text{IL}} + n_{\text{IL dark}}^2 + n_{\text{OOL dark}}^2 + n_{\text{loop}}^2 + n_{\text{analyser}}^2} \quad (1.3)$$

where  $q$  is the charge of the electron,  $I_{\text{OOL/IL}}$  is the current through the out-of-loop and in-loop photodiodes,  $n_{\text{OOL/IL dark}}$  is the dark noise of the out-of-loop photodiodes and in-loop photodiode,  $n_{\text{loop}}$  is the electronic noise of the loop and  $n_{\text{analyser}}$  is the noise of the analyser.

The light incident onto the FD10D was amplitude modulated at 1111 Hz. This modulation was used to calibrate the measurements of noise as the relative intensity of this peak was known and could be checked against with the signal measured by the high performing in-loop photodiode. By tracking the height of an intensity modulation in the light for a series of powers up to 16 mW, the saturation limit of the photodiode could be investigated.

### 1.5.2 Results

The effect of increasing the reverse bias on the quantum efficiency of the FD10D is shown in Figure 1.2. For all beam intensities that the photodiode was exposed to, the increase in quantum efficiency was  $\sim 15\%$  as the reverse bias went from 0 V to 1.8 V, the stated maximum reverse bias in the FD10D's datasheet. There may be a saturation related effect at the 1% level, beams with a lower intensity experienced a greater increase in  $\eta$  than beams with a higher intensity, however the quantum efficiency may not be uniform over the photodiode – the spatial dependence of  $\eta$  can be crater-like with  $\eta$  being greatest at the edges of the photodiode. The difference between the DC and AC response could only be described by a non-linear process. These effects are small and Figure 1.2 gives a clear incentive to increase the bias voltage to improve the quantum efficiency of the FD10D.

Figure 1.3 shows the photocurrent measured has the noise level predicted by Equation 1.3. This means that the increase in photocurrent at a reverse bias voltage of 0.6V is, without doubt, an increase in quantum efficiency. At 0.6 V reverse bias, for DC photocurrents ranging from 7.5 mA to 15.5 mA, the calibration peak and the noise level of the photocurrent were measured to ensure the photodiode was not saturating. Over this range there was a 2% change in the ratio between the DC and AC measurement, this is shown in Figure 1.4. The SNR of this measurement is limited to the 1% level due to the difference between the calibration peak and the level of shot noise.

Taking into account the light reflected by the FD10D's window, at 0.6 V reverse bias, the quantum efficiency was calibrated using two other power meters. The

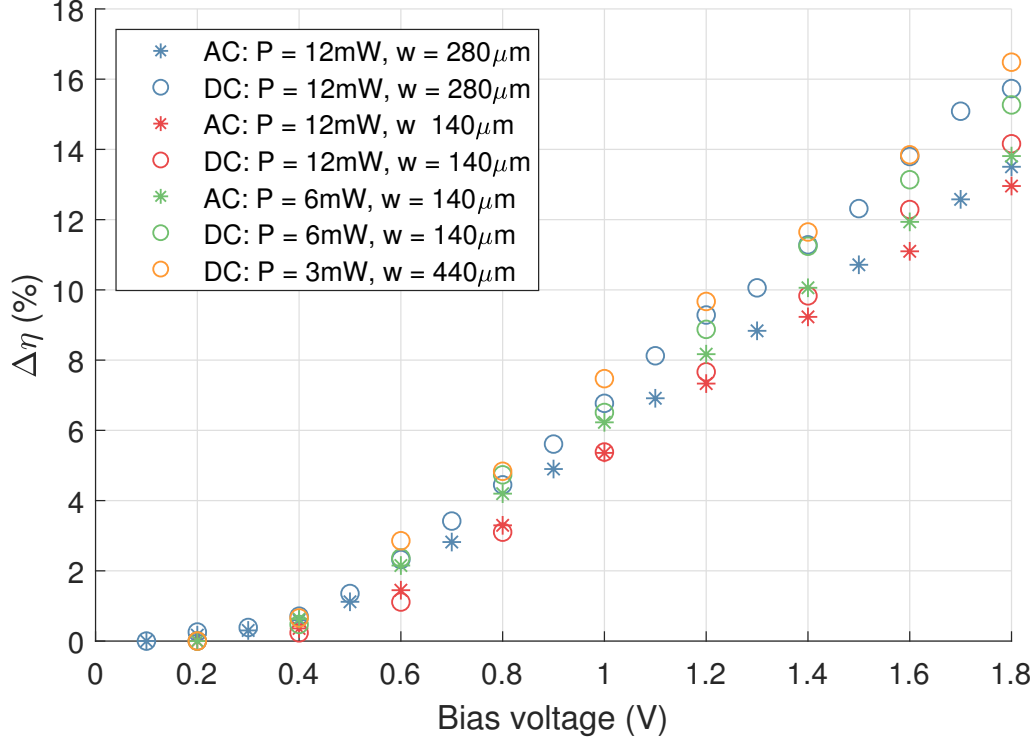


Figure 1.2: The FD10D’s quantum efficiency depends on its reverse bias; the increase in quantum efficiency was at the  $\sim 15\%$  level when the bias was increased to its maximum datasheet value. Two types of measurement were used to determine the increase in quantum efficiency as the bias was increased. The first method was to measure the DC response and the corresponding results are shown as circles. The second method involved modulating the bias and measuring the resulting modulation in the photocurrent, i.e. an AC measurement, and these are shown by the stars. Each set of measurements was done with a different intensity by changing either the beam’s width or power.

internal quantum efficiency was measured to be 70%. There is a significant amount of light reflected by the FD10D's window; if the window is not removed, at 0.6 V reverse bias the quantum efficiency would be 60%.

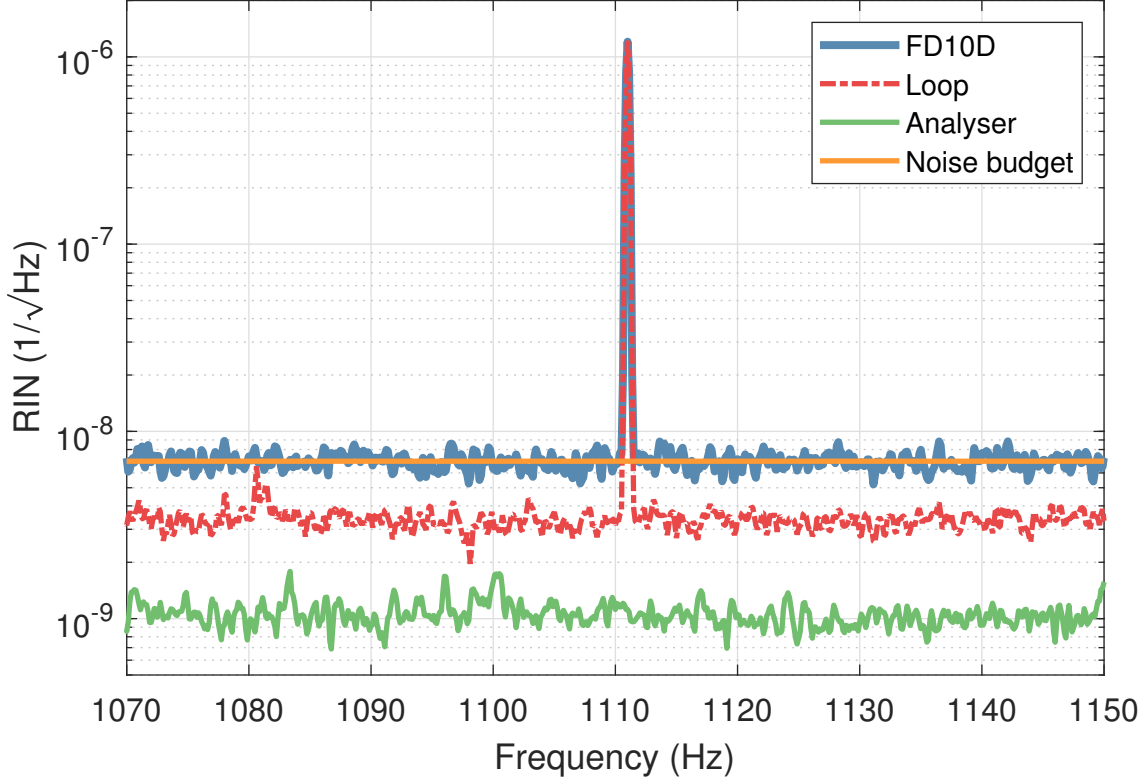


Figure 1.3: The bias of the FD10D was set to 0.6 V and it was exposed to light [find values](#). The noise of the photocurrent produced by the FD10D (blue line) corresponds to the Equation 1.3 (orange line) meaning that that the quantum efficiency of the FD10D did increase rather than the increase in photocurrent being the result of an avalanche process. The 1111 Hz signal was used to calibrate the measurement.

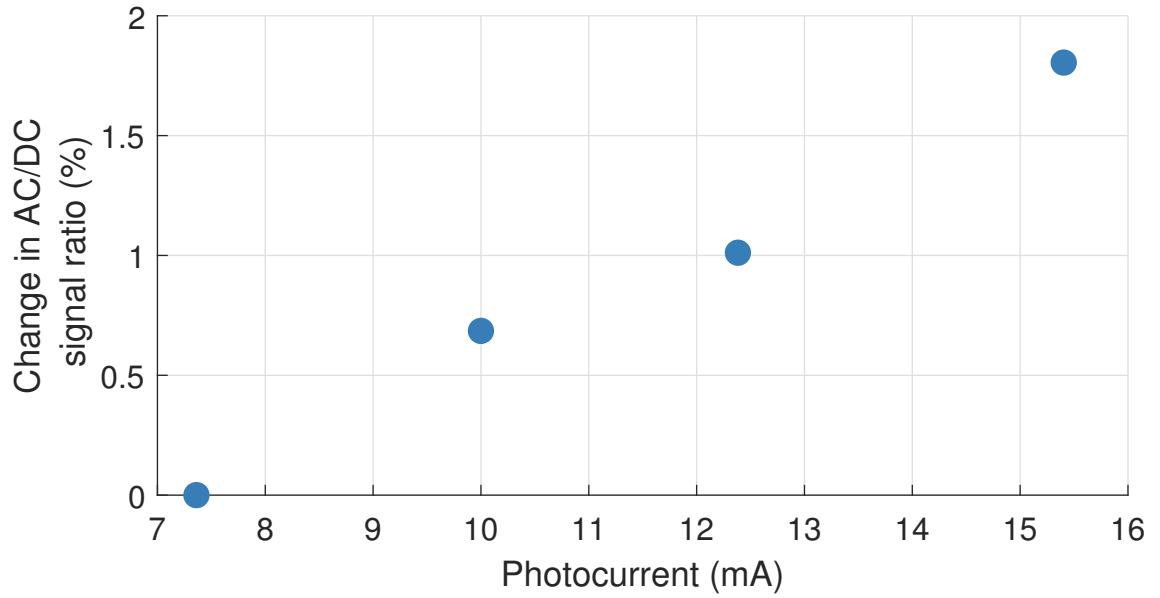


Figure 1.4: The bias of the FD10D was set to 0.6 V and it was exposed to a series of different power levels to check at what photocurrent the photodiode would saturate at. This was done by measuring the height of the calibration peak; if this peak decreased in size relative to the DC signal then this would indicate the photodiode was saturating. This peak could only be measured to within 1%, and so the FD10D showed no sign of saturation under these conditions.

## 1.6 Dark Current and Dark Noise in Extended InGaAs Photodiodes

The dark noise of the extended InGaAs photodiode used in previous quantum optics experiments [4,5] resulted in the photodiode being run without a bias, thus the dark noise of the photodiode was a limiting factor in these experiments. The dark current and dark noise of extended InGaAs photodiodes depends on their reverse bias. The larger the reverse bias, the higher the dark current and dark noise.

There is a balance to strike between the quantum efficiency and dark noise. It is desirable to have as large a reverse bias as possible as this increases the quantum efficiency and saturation limit of the photodiode (see Section ?? or Figure ??). The dark noise of the photodiode should be below the shot noise level of the light that will be detected. As discussed in Section ??, the shot noise of a 10 mA photocurrent was used as a bench mark to compare the dark noise of the photodiodes at 10 Hz to.

To measure the dark noise produced by the extended InGaAs photodiodes, a transimpedance amplifier with high gain was used. This circuit, shown in Section B.1, allows for measurements of current noise several hundred times smaller than the shot noise of a 10 mA photocurrent ( $6 \times 10^{-11} \text{ A}_{\text{rms}}/\sqrt{\text{Hz}}$ ). The photodiodes were placed in a light tight box while the measurement was being made.

### **1.6.1 The Dependence of Dark Noise and Dark Current on Reverse Bias in Extended InGaAs Photodiodes**

The dark noise as a function of bias was measured at room temperature ( $\sim 20^\circ\text{C}$ ) for the photodiodes listed in Table 1.1 . The results for the FD10D are shown in Figure 1.5. The dark noise for the rest of the photodiodes can be found in Appendix A.1. Not all photodiodes had a simple noise spectrum; however, usually, the spectrum would fall as  $1/f^{0.5}$ .

The dark noise at 10 Hz and the dark current as a function of bias for each photodiode is shown in Figure 1.6. The noise at 10 Hz was either determined by reading the value from the corresponding graph in Appendix A.1 or, if more precision was required, by fitting a  $1/f^\beta$  slope to the data and inferring the noise from the fitted line.

Where there were duplicate photodiodes to choose from, the superior photodiode was selected for characterisation. There can be a significant variation in quality within a batch of photodiodes. Figure 1.7 shows there was a difference of two orders of magnitude in the dark noise between two ‘identical’ Hamamatsu photodiodes.

### **1.6.2 The Dependence of Dark Noise and Dark Current on Temperature in Extended InGaAs Photodiodes**

The dark noise of extended InGaAs photodiodes can be reduced by cooling them. The typical low-temperature damage threshold for extended InGaAs photodiodes is  $-40^\circ\text{C}$ , so the dark noise of the photodiodes listed in Table 1.1 was measured at a

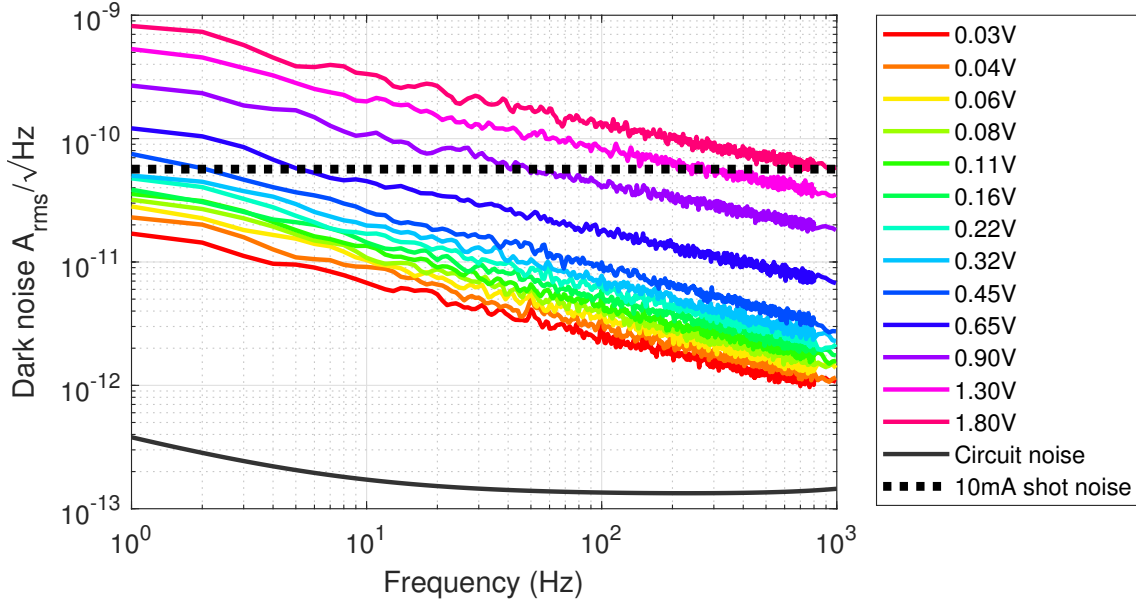


Figure 1.5: The dark noise of the FD10D increases as the reverse bias is increased. Above 0.6 V, the dark noise of the FD10D would limit the noise of a 10 mA current above 10 Hz. As the noise spectral densities have no knees, the function  $i_c/f^\beta$  can be fitted to the it. See appendix A.1 for the dark noise spectra of other photodiodes. Below 0.9 V reverse bias, the spectra fall as  $f^{-0.45}$  and above 0.9 V bias the spectra fall as  $f^{-0.4}$ . This is typical for ‘ $1/f$ ’ noise. This data is shown along side the results for the other photodiodes tested in Figure 1.6.

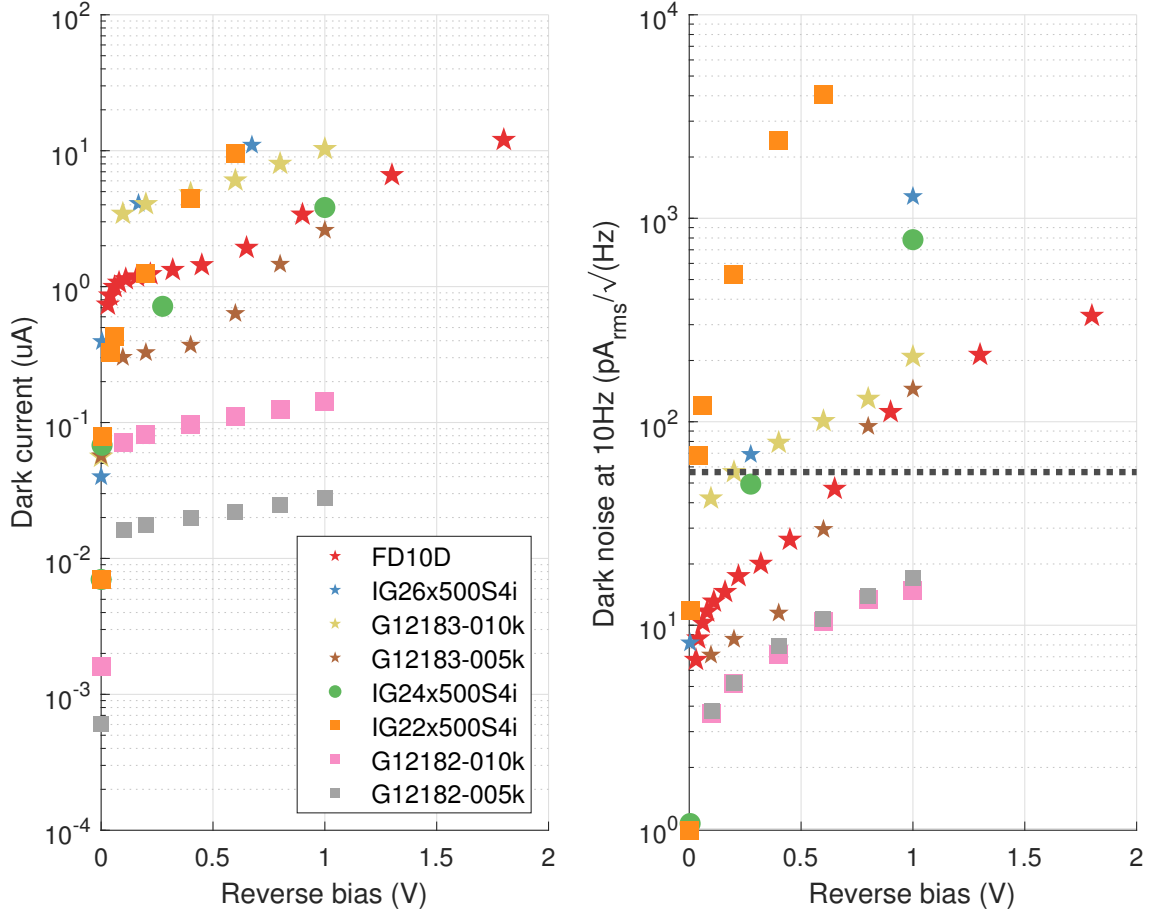


Figure 1.6: Left panel: the dark current for each photodiode was measured. Photodiodes of a similar cut-off wavelength share the same marker shape. The 1 mm photodiodes have larger markers than the 0.5 mm photodiodes. Right panel: the dark noise at 10 Hz for each photodiode. The dashed line indicates the level of shot noise for a 10 mA current.

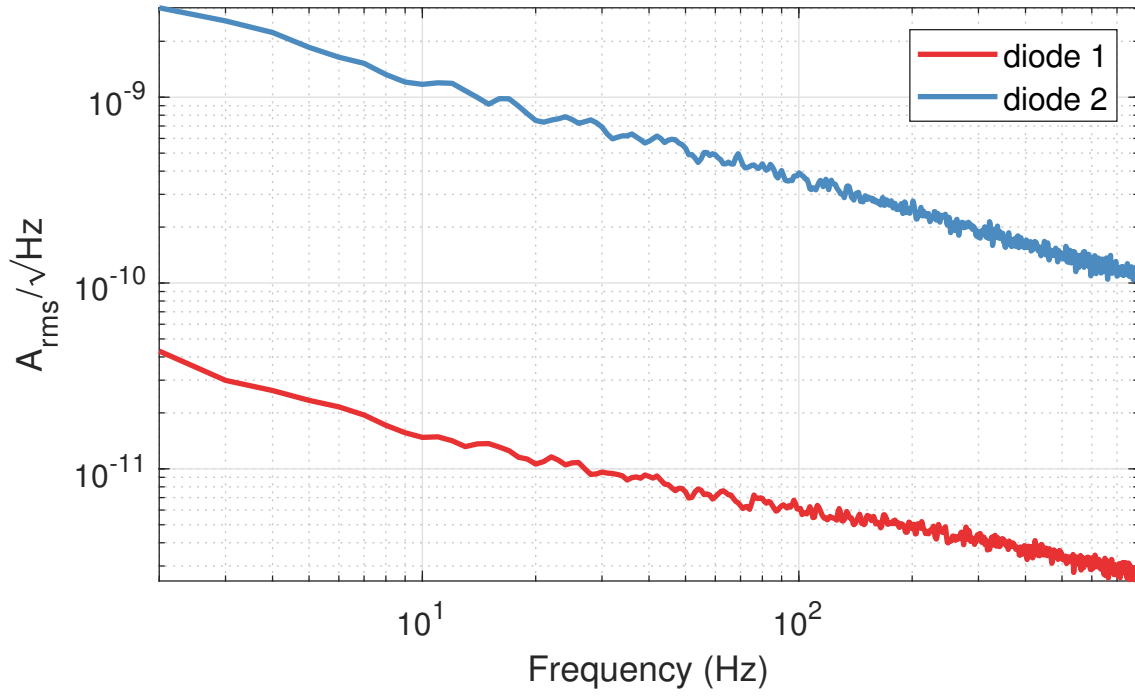


Figure 1.7: The difference in dark noise between two Hamamatsu photodiodes (g12182-010k) of the same kind under identical operating conditions (room temperature at 1 V bias). Photodiode 2 performs nearly 100 times better than photodiode 1.

range of temperatures from  $-40^{\circ}\text{C}$  to  $30^{\circ}\text{C}$ .

The temperature of the photodiode was controlled with power resistors and an aluminium block thermally connected to a liquid nitrogen bath. The block is shown in Figure 1.8. This block had a copper rod soldered to it and this rod was inserted into the liquid nitrogen bath. A computer-controlled power supply (Temna 72-2540) was connected to the power resistors. A pt1000 temperature sensor and a multimeter (Keithley 2000), configured to make a four wire resistance measurement, were used to measure the temperature of the block. Labview was used to perform the PID calculation. This program read in the temperature of the block and calculated the voltage to set the power supply to.

The dark noise and dark current for each photodiode in Table 1.1 was measured at the maximum reverse bias specified in its datasheet. The dark noise at 10 Hz and dark current for each photodiode is shown in Figure 1.9. The individual dark noise spectra for each photodiode at each temperature is shown in Appendix A.2. Except for the IG24x500S4i, the dark noise and dark current for each photodiode fell exponentially as the photodiode was cooled. The IG24x500S4i had a knee-like feature in its spectrum. Above 10 Hz, the dark noise fell as the IG24x500S4i was cooled, however below 10 Hz the noise spectra were bunched together and had a weak dependence on temperature.

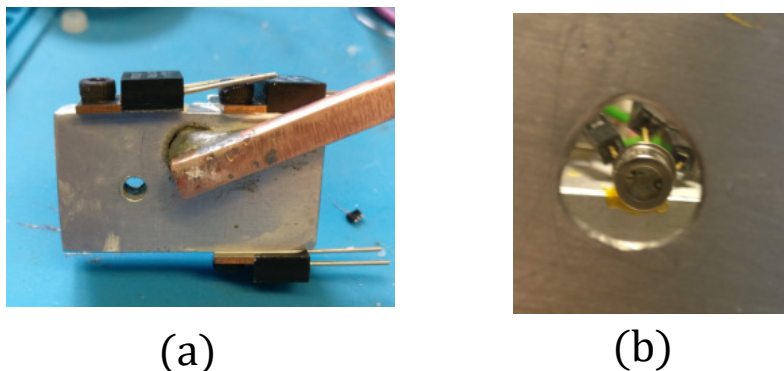


Figure 1.8: (a) The photodiode was affixed to this aluminium block. The power resistors provided heating to the block and the copper rod provided a thermal link between the liquid nitrogen bath and the block. (b) A photodiode glued to the block such that a large proportion of the photodiode was in contact with it.

## 1.7 The Saturation Limit of Some Laser Components Extended InGaAs Photodiodes

The laser components photodiodes were found to saturate at low powers if they were biased with a low enough voltages such that their dark noise was below the noise target. The laser components photodiodes were exposed to a light modulation of known size. A similar type of measurement is reported in [24]. The size of the light modulation could be cross-checked by measuring the it on the high performance in-loop photodiode. The power of the light was increased, and the decrease in the relative intensity of the AC modulation was recorded. This is shown in Figure 1.10. A decrease in AC signal indicates that the photodiode is saturating.

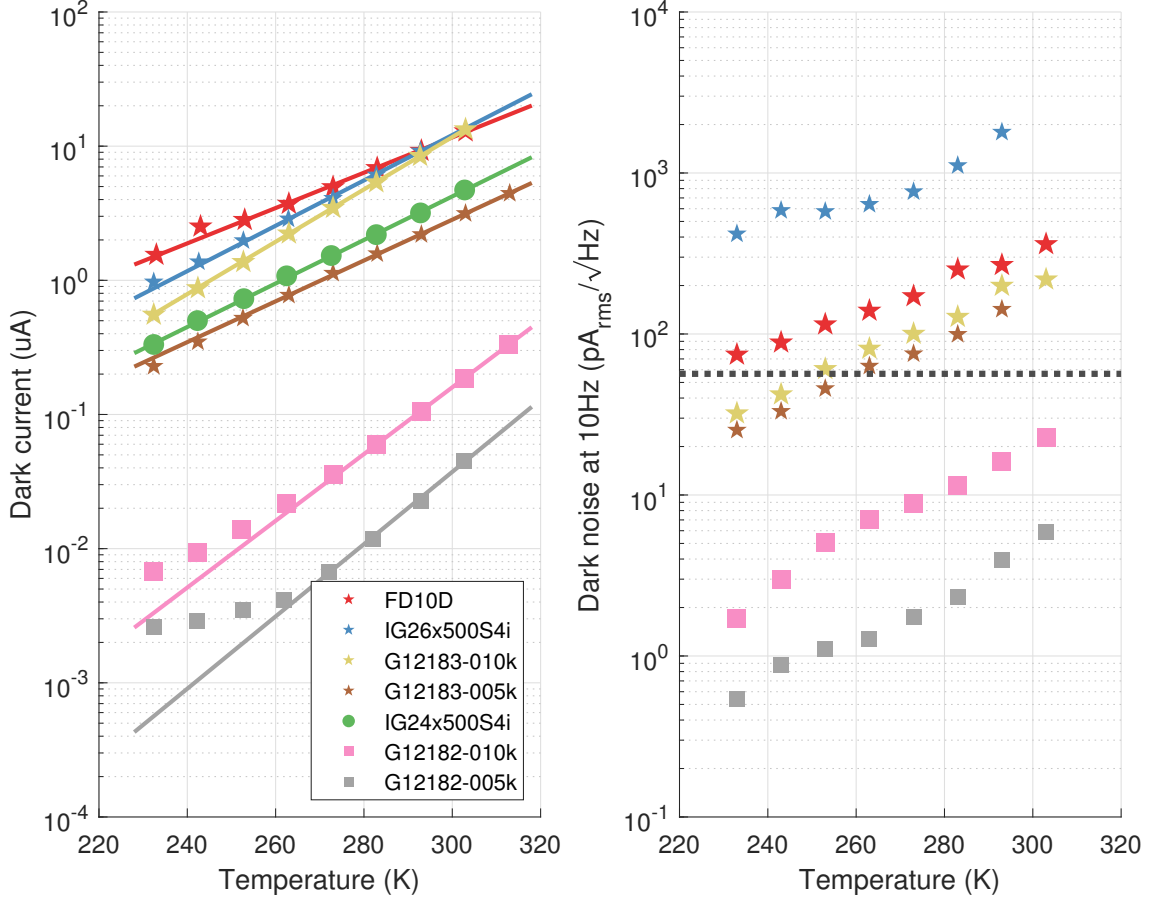


Figure 1.9: Left panel: the dark current for each photodiode at their maximum datasheet specified reverse bias as a function of temperature. The apparatus limited dark current measurements to a couple of nanoamps. Exponentials were fitted to the data. Photodiodes with similar cut-off wavelengths share marker shapes, and 1 mm photodiodes have larger markers than the 0.5 mm ones. Right panel: the dark noise of each photodiode was measured as a function of temperature. The black dashed line indicates the shot noise level for a 10 mA current. At 10 Hz, the measurements for the IG24x500S4i did not change much as the temperature changed, there was a knee around this frequency (see Figure A.13); the noise of the IG24x500S4i was  $600 - 700 \text{ pA}/\sqrt{\text{Hz}}$  at 10 Hz.

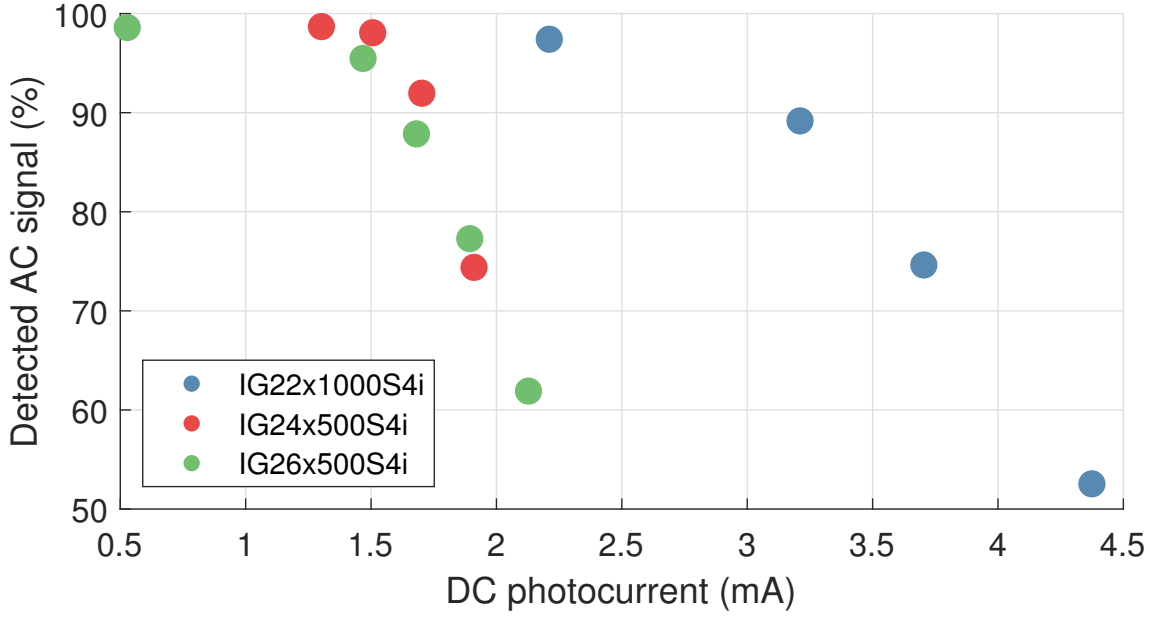


Figure 1.10: The saturation limit of three Laser Components photodiodes was measured. The photodiodes were biased so that their noise at 10 Hz was equal to the shot noise of a 10 mA current. The IG24x500S4i and IG26x500S4i were exposed to a beam whose  $3\sigma$  width was 220  $\mu\text{m}$  and the IG22x1000S4i was exposed to a beam whose  $3\sigma$  width was 1000  $\mu\text{m}$ . The size of the light modulation whose relative size was known was tracked as the DC photocurrent was increased. As the photodiodes saturated, the relative size of the AC signal decreased. These photodiodes had low saturation limits compared to the FD10D, which showed no signs of saturation up to 16.5 mA (see Figure ??).

## 1.8 Excess Noise in the Photocurrent made by the IG24x500S4i

The noise of the photocurrent produced by the IG24x500S4i exceeded shot noise by a factor of  $\sim 2$ . The set-up for this experiment is explained in Section 1.5. The photodiode was not saturating during this measurement as the 1111 Hz peak had the right relative intensity. This excess noise was found at 1.3 mA and 0.25 mA of photocurrent.

To rule out bias noise causing the excess noise, a noise projection measurement was made. The transfer function between the voltage at the anode of the photodiode and the output of the transimpedance amplifier was measured while 1 mA of photocurrent was being generated by the photodiode. A calibration modulation was applied to the bias voltage, and the noise at the anode of the photodiode and the noise at the output of the transimpedance amplifier were simultaneously measured. Using this method, 42.5 dB of bias voltage noise was removed from the measurement; however, this did not remove the excess broadband noise. This is shown in Figure 1.12.

The ideality factor of the IG24 was investigated. The ideality factor differing significantly from 1 indicates that additional behaviour that was not described in Section 1.2 occurring within the photodiode. The equation of a photodiode is

$$I = I_p + I_s \left( \exp \left( -\frac{qV_b}{nk_B T} \right) - 1 \right), \quad (1.4)$$

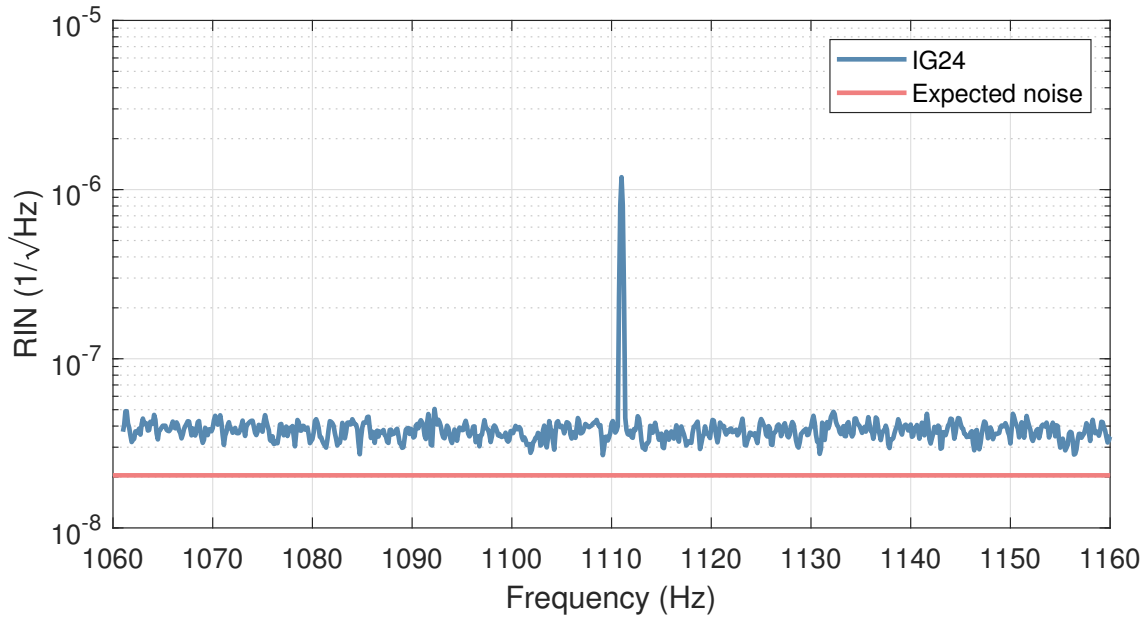


Figure 1.11: The noise of a 1.3 mA of photocurrent made by the IG24x500S4i (blue) exceeded the expected amount (red) based on Equation 1.3 by a factor of 1.75. This beam was not saturating the photodiode as the height of the 1111 Hz signal was the right level.

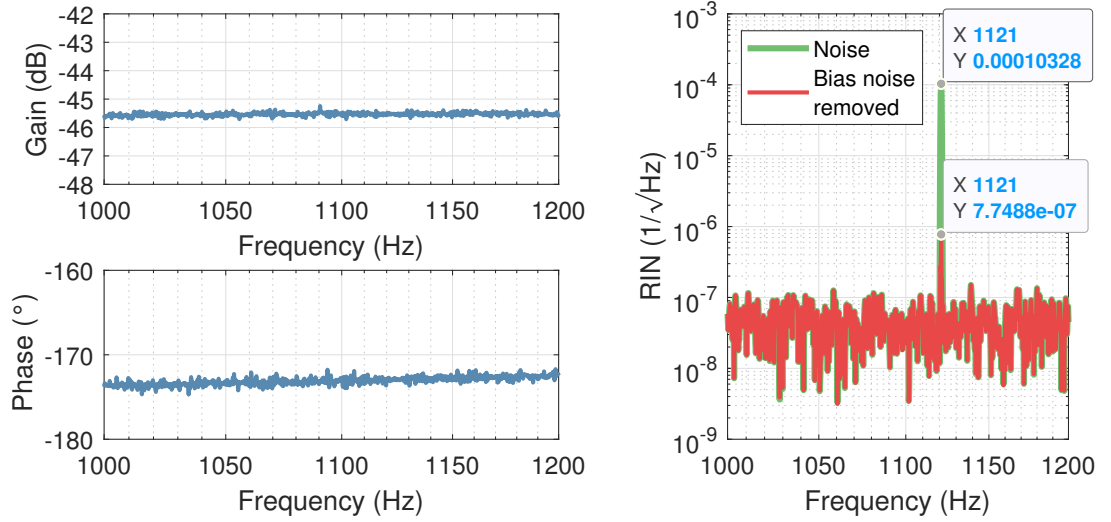


Figure 1.12: A noise projection measurement was performed to determine if the excess noise in the photocurrent produced by the IG24x500S4i (see Figure 1.11) was due to bias noise. The left hand panels show the bode plot of the transfer function between the photodiode's cathode and the output of the transimpedance amplifier when the photodiode was producing 1.3 mA of photocurrent. The right hand panel shows the effect of subtracting bias noise from the signal at the output of the transimpedance amplifier. The peak at 1121 Hz was a modulation applied to the photodiode's cathode; 42.5 dB of this signal was removed by using the noise projection technique which is in close agreement with the amount expected from the measured transfer function. However, the level of the broadband noise remained unchanged thus indicating that the excess noise witnessed in the photocurrent produced by the IG24x500S4i was not due to bias noise.

where  $I_p$  is the photocurrent,  $I_s$  is the saturation current of the photodiode,  $q$  is the charge of the electron,  $V_b$  is the bias of the photodiode,  $n$  is the ideality factor,  $k_B$  is Boltzmann's constant and  $T$  is the temperature of the photodiode. The current flowing through the photodiode as a function of forward bias was measured and is shown in Figure 1.13. The SciPy python function `scipy.optimize.curve_fit` was used to do an unweighted fit of the data to Equation 1.4. The photodiode was found to have  $n \simeq 1$ . An additional check of Equation 1.4 was made by shining light was shone onto the photodiode. This is shown in Figure 1.14.

The source of this excess noise is still unclear: the photodiode was not saturating as the calibration signal was the right level relative to the DC photocurrent, a noise projection measurement showed that the bias voltage noise was not the source of the excess photocurrent noise, and a measurement of the ideality factor was used to determine if the photodiode behaved in as an ideal photodiode would. The level of noise cannot be explained by the Johnson-Nyquist noise of the shunt resistance of the photodiode as the excess noise is far too high.

## 1.9 Properties of an InSb detector

InSb detectors have high quantum efficiency [6] to short infrared wavelength light. A P5968 detector was tested, and the noise of the current produced by it was investigated. InSb detectors must be operated at low temperatures so liquid nitrogen was used to cool the photodiode to 77 K. The transimpedance amplifier used to measure the photocurrent is shown in Appendix B.1.

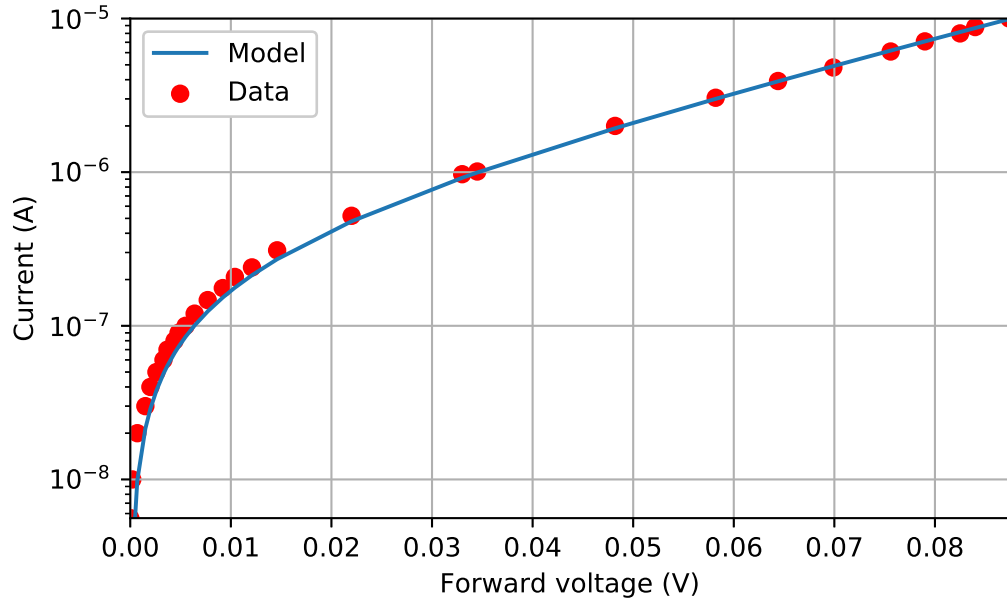


Figure 1.13: To investigate possible reasons why there was excess noise in the photocurrent produced by the IG24x500S4i, its ideality factor was measured (red points). Equation 1.4 was used to do an unweighted fit to the data ( $I_0 = (360 \pm 10) \text{ nA}$  and  $n = 1.02 \pm 0.01$ ). The IG24x500S4i was found to be close to ideal as  $n$  was close to 1, and so this does not provide clues to the source of excess noise in the photocurrent.

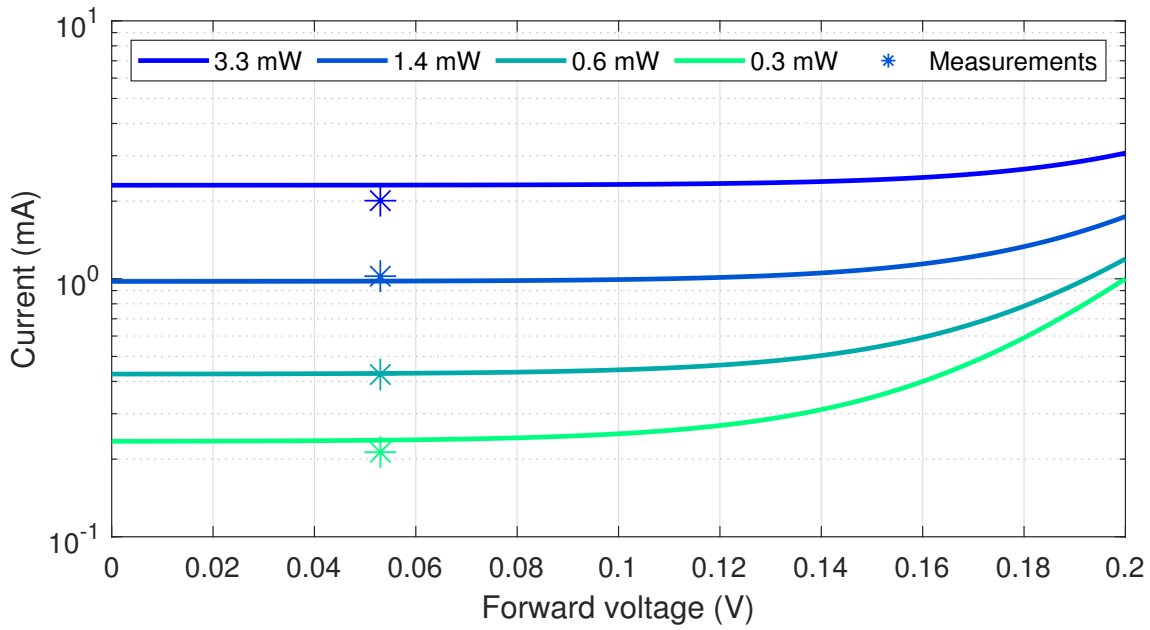


Figure 1.14: Different light levels were shone onto the photodiode and the current produced at 0.05 V reverse bias was recorded. These measurements are shown by the stars. Based on the model obtained from the data shown in Figure 1.13, the current flowing through the photodiode was predicted (indicated by the lines). The model and the measurements are in agreement with each other although the photodiode may have been saturating for the 3.3 mW measurement.

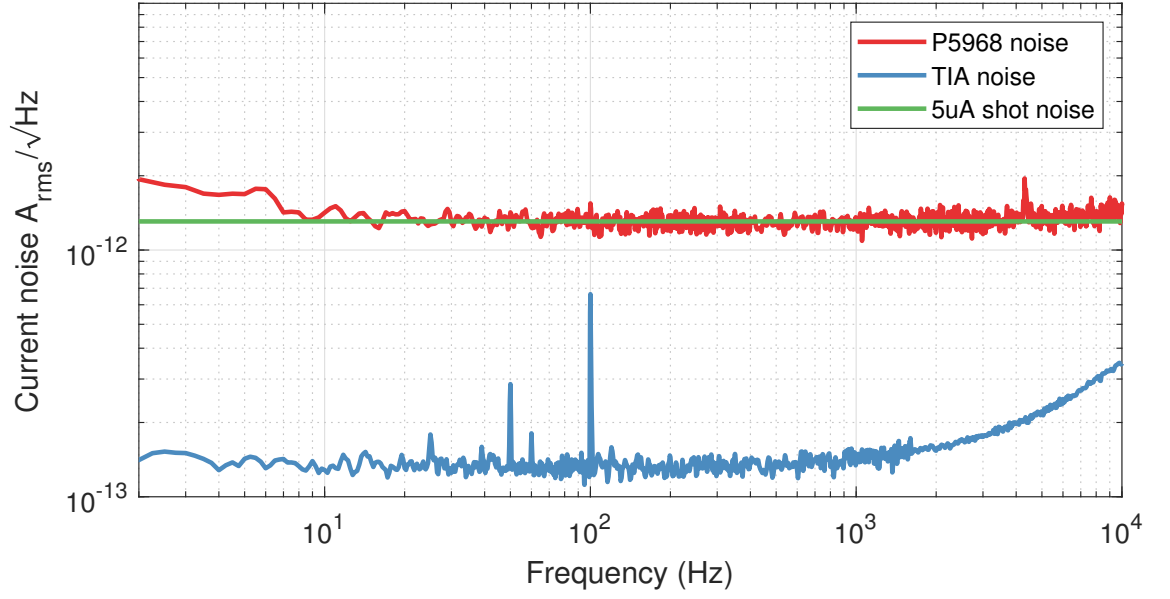


Figure 1.15: The P5968 InSb detector had its cathode grounded and the noise of the photocurrent due to the blackbody radiation produced by room temperature aluminium was measured (red). The expected level of photocurrent was calculated to be  $5.5\text{ }\mu\text{A}$  and the measured photocurrent was  $5.3\text{ }\mu\text{A}$ . The noise level of the photocurrent corresponds to shot noise (green). The noise of the transimpedance amplifier (blue) is well below the shot noise of the photocurrent.

The detector was characterised at by placing a low emissivity material in front of the detector. The InSb is sensitive to wavelengths of up to  $5.3\text{ }\mu\text{m}$ . Based on the solid angle covered by the material as seen by the detector, the expected level of thermal radiation that would be detected could be calculated. This expected photocurrent was  $5.5\text{ }\mu\text{A}$ . The DC current from the device was measured to be  $5.3\text{ }\mu\text{A}$ . The results from this test are shown in Figure 1.15. This corresponds to the device having a quantum efficiency of  $\sim 90\%$ .

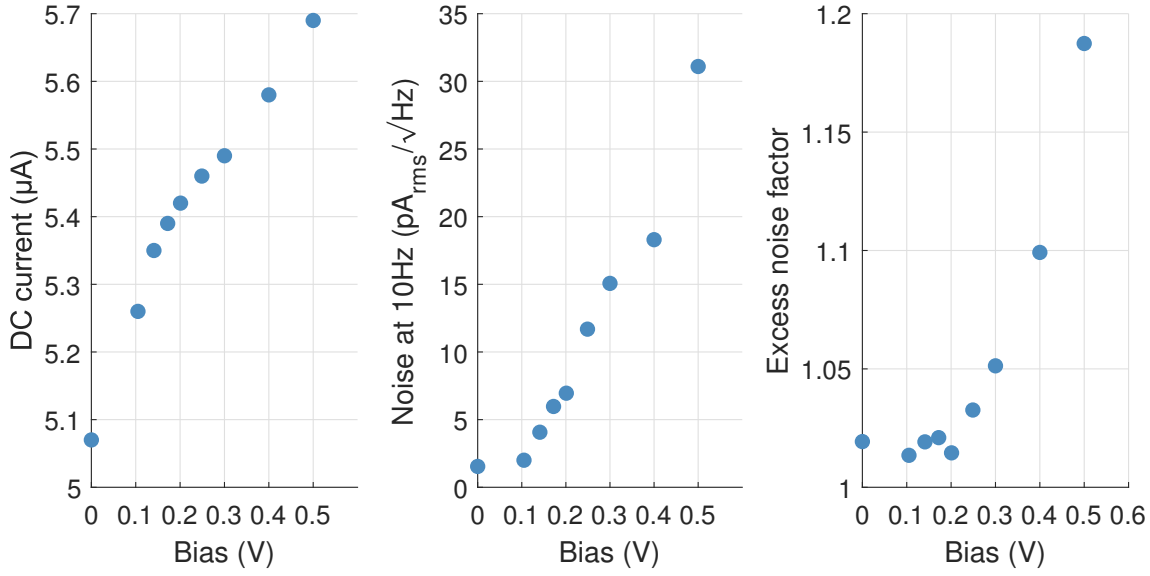


Figure 1.16: Left panel: the DC dark current of the P5968 as a function of bias. Middle panel:  $1/f$  noise rose as the bias of the P5968 was increased (see Figure A.15). Right panel: At a frequency where  $1/f$  was not present, the noise of the photocurrent was measured. The excess noise factor is the multiplicative factor that the shot noise corresponding to the DC photocurrent needs to be multiplied by to get the noise level that was measured.

As the quantum efficiency of InSb photodetectors depends strongly on the reverse bias for  $1.55\mu\text{m}$  wavelengths [25], the behaviour of the P5968 under a reverse bias was investigated. A reverse bias was applied to the device and this generated  $1/f$  noise. The full data is shown in Section A.3. The noise of the photocurrent also showed a broadband increase, this is shown in Figure 1.16.

## 1.10 Discussion

None of the photodiodes tested in this Chapter would meet the requirements described in Section 1.3 as the quantum efficiency is specified in all of their datasheets to be below 99%. Communication with several commercial manufacturers of extended InGaAs photodiodes suggest that there is no off-the-shelf photodiode capable of meeting the requirements set out in Section ??; the results in Section 1.5 and 1.6 support and give insights into the best possible operating conditions for an off-the-shelf extended InGaAs photodiodes.

The results cover seven different models of extended InGaAs photodiode and so it is possible that another photodiode manufacturer (e.g Teledyne Judson) have better performing photodiodes than the ones tested in this chapter. Additionally, as only two copies of most photodiodes were available for testing, batch-level statistics cannot be obtained from the results. For instance, the FD10D reported on in this chapter has lower dark noise than the G12183-010k; however, this does not tell one if FD10Ds are generally better than G12183-010ks.

Section 1.5 shows the effect that biasing an extended InGaAs photodiode can have on its quantum efficiency. The quantum efficiency of the FD10D was increased by 15% when its bias was set to its maximum value. This increase in quantum efficiency occurs because the electron-hole pairs that are excited by the incoming photons spend less time within the depletion region, thus they have a smaller chance of recombining with another hole/electron. This motivates using as large a bias as possible as the quantum efficiency can be greatly increased.

The high amount of dark noise associated with the bias that was found in the Laser Components range of photodiodes (see Section??) meant that they had to be operated with low biases if they were to be able to produce shot noise limited photocurrents above 10 Hz. This resulted in them having a low saturation level, see Figure 1.10. This is another reason why biasing a photodiode may be necessary. To avoid this saturation limit, one could split the beam and use an array of photodiodes to detect the light. However, better performing photodiodes could be used instead; the FD10D showed no signs of saturation, see Figure 1.4, when operating with a bias that fulfilled the noise criteria.

Usually, applying a reverse bias is done to increase the bandwidth of a photodiode, however a few hundred picofarads of capacitance is a manageable amount to design a transimpedance amplifier around for a DC readout scheme such as that used in the LIGO detectors. For instance, the photodiode circuit used in Section 1.5 (see Appendix B.2) was quiet enough to be limited by the shot noise of a 10 mA photocurrent.

While the dark current does not depend strongly on the reverse bias when the bias is below the tunnelling threshold, the dark noise of every photodiode increased as its reverse bias increased. Although photodiodes with higher dark current tended to have higher dark noise, the dark noise could not be predicted from the dark current. For instance, the FD10D had a larger dark current but lower noise than the IG24x500S4i. The dark current tended to be greater in larger photodiodes, this is due to larger photodiodes being more likely to contain defects. However, the Laser Components photodiodes break this rule as they were all of low quality. The

Hamamatsu photodiodes with shorter cut-off wavelengths (less indium), had lower dark current and dark noise compared to the Hamamatsu photodiodes with longer cut-off wavelengths.

The dark noise can vary considerably amongst a batch of photodiodes, for example there was nearly a factor of 100 between the dark noise of the two Hamamatsu G12182-010k that were tested (see Figure 1.7). Whether this is due to their manufacturing or how they were handled prior to purchase is unclear, however this result shows that it may be worth buying a batch of Extended InGaAs photodiodes and selecting the best ones from it.

The dark noise of a photodiode can be significantly reduced by cooling it. The dark current and dark noise of a photodiode increases exponentially with temperature. Photodiodes can be purchased with thermoelectric coolers, however integration of such a photodiode into a vacuum system was not investigated and could pose an engineering challenge.

## 1.11 Conclusion

Extended InGaAs photodiodes are widely available detectors that can be used to sense  $2\mu\text{m}$  light; however, off-the-shelf detectors do not fulfil the requirements of a photodiode that can be used in the readout of a gravitational wave detector. This is due to gravitational wave detectors needing high quantum efficiency photodiodes so that the shot noise of the signal is not significantly enhanced.

It is technically challenging to create Extended InGaAs photodiodes with high quantum efficiency due to the lattice mismatch between the InP substrate and the InGaAs. While the depletion region of a photodiode can be made larger by running the photodiode with a reverse bias, the dark noise of an Extended InGaAs photodiode increases when reverse biased. Therefore, dark noise measurements of a selection of photodiodes (see Figure 1.9) were made to investigate suitable operating conditions for them.

To overcome the manufacturing challenges associated with creating extended InGaAs photodiodes, a dedicated wafer would need to be commissioned. Increases in quantum efficiency can be obtained by fabricating photodiodes to have anti-reflective coatings; however, the challenge of creating high quantum efficiency photodiodes lies in making is thick enough to absorb  $> 99\%$  of the incident photons. Additionally, the photodiode must have a low amount of defects if it is to have a quantum efficiency  $> 99\%$ . Reference [26] shows results from a photodiode with 95% quantum efficiency and a cut-off wavelength of  $2.4\text{ }\mu\text{m}$ , however due to the exponential dependence of absorption on thickness, it is still a significant challenge to reach  $> 99\%$  quantum efficiency. Future work may include investigating the growth conditions for extended InGaAs photodiodes as it has been shown that the dark noise of a photodiode made from extended InGaAs can be reduced by orders of magnitudes by optimising this.

InSb detectors may also be useful as they have near unity quantum efficiency at short infra-red wavelengths, however their quantum efficiency below  $2\text{ }\mu\text{m}$  can exceed unity due to multi-electron processes [6]. An InSb device was tested, however its

characteristics at  $2\text{ }\mu\text{m}$  are still unclear. To investigate these the device's window would need removing and a  $2\text{ }\mu\text{m}$  laser would be needed.

Alternative photodiode materials could be used to detect  $2\text{ }\mu\text{m}$  light. There have been suggestions that HgCdTe could be used to make a photodiode with  $> 99\%$  quantum efficiency at  $2\text{ }\mu\text{m}$  as HgCdTe does not suffer from strain related problems, however the linearity and dark noise of such HgCdTe photodiodes is unknown.

# Appendix A

## Extended InGaAs Diodes Noise Spectrums

### A.1 Bias Section

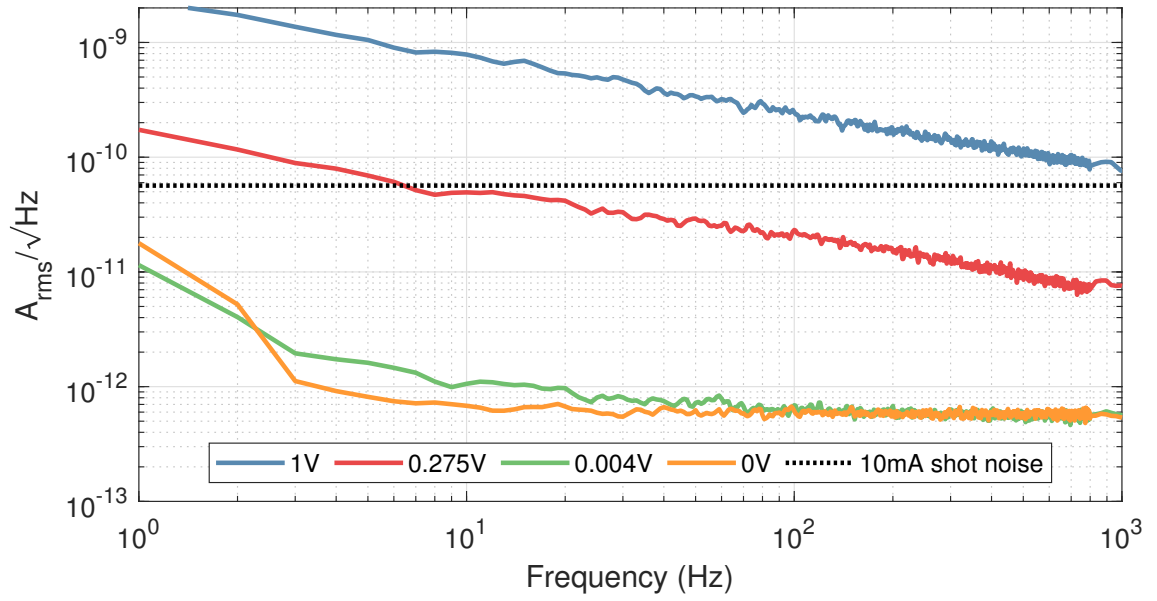


Figure A.1: IG24

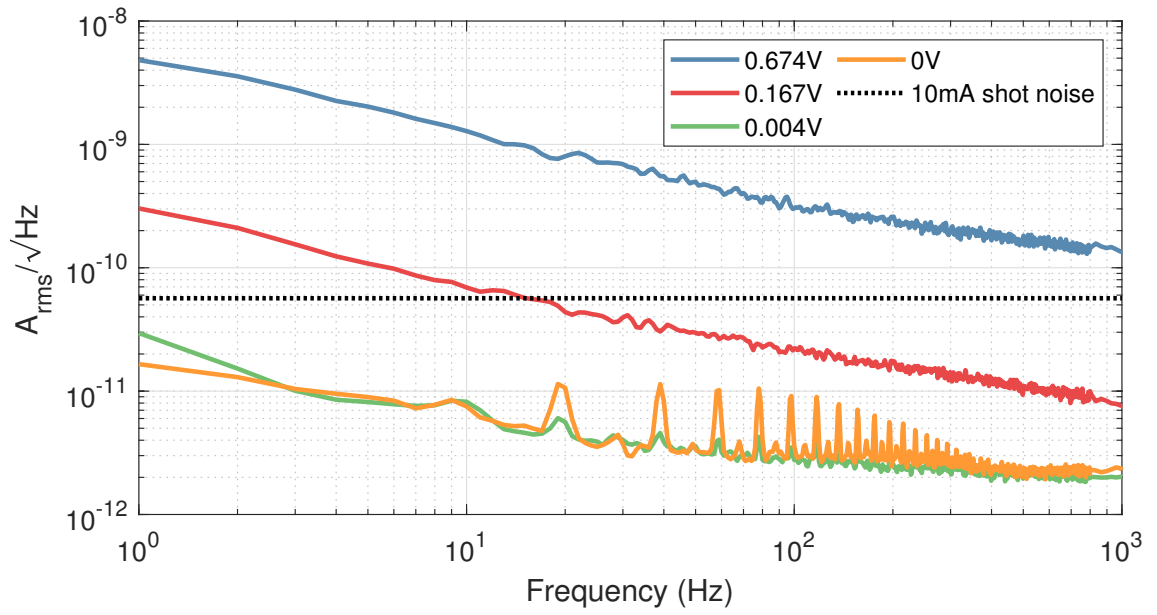


Figure A.2: IG26

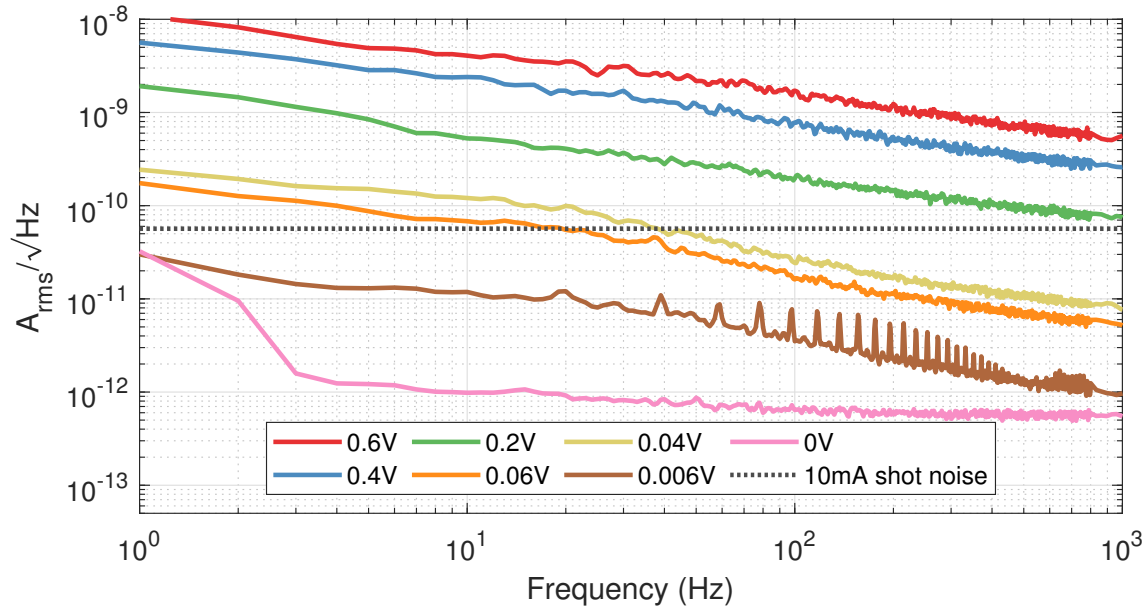


Figure A.3: IG22

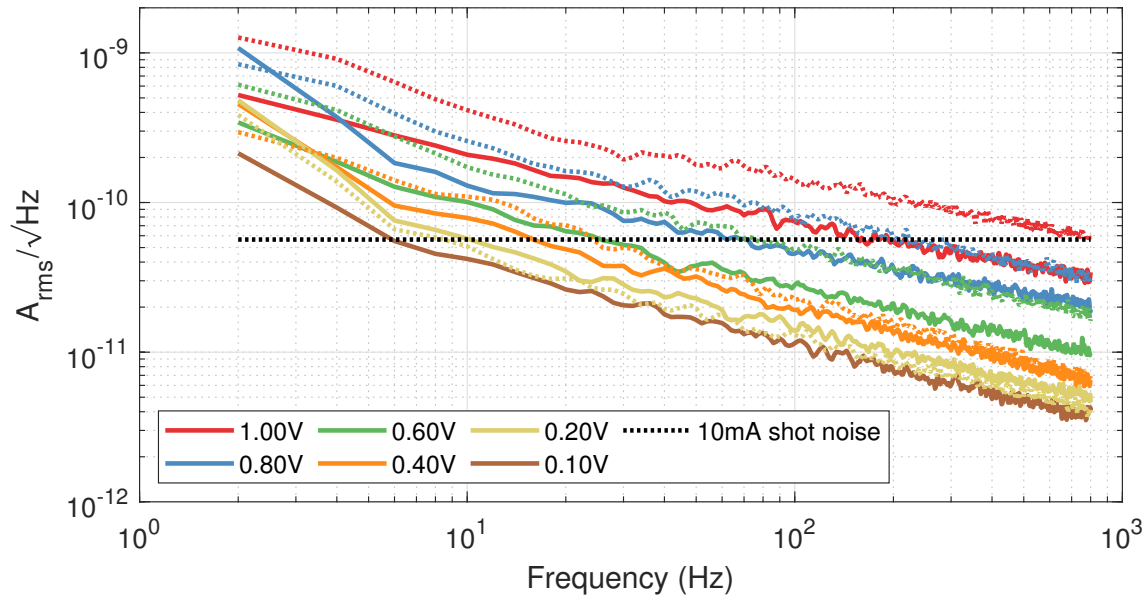


Figure A.4: G12183-010k

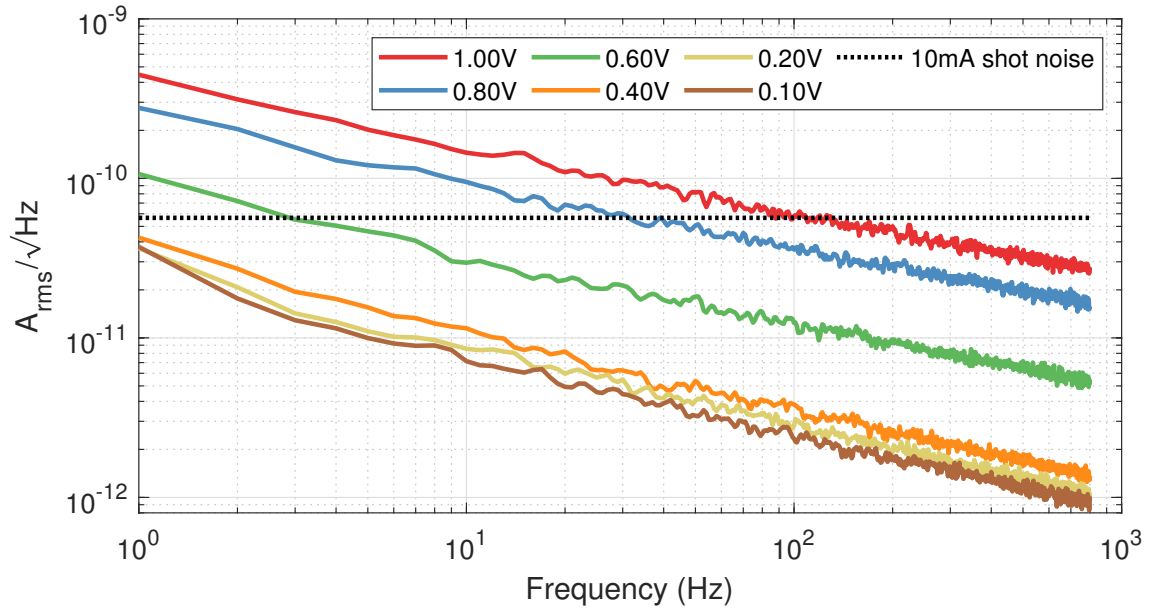


Figure A.5: G12183-005k

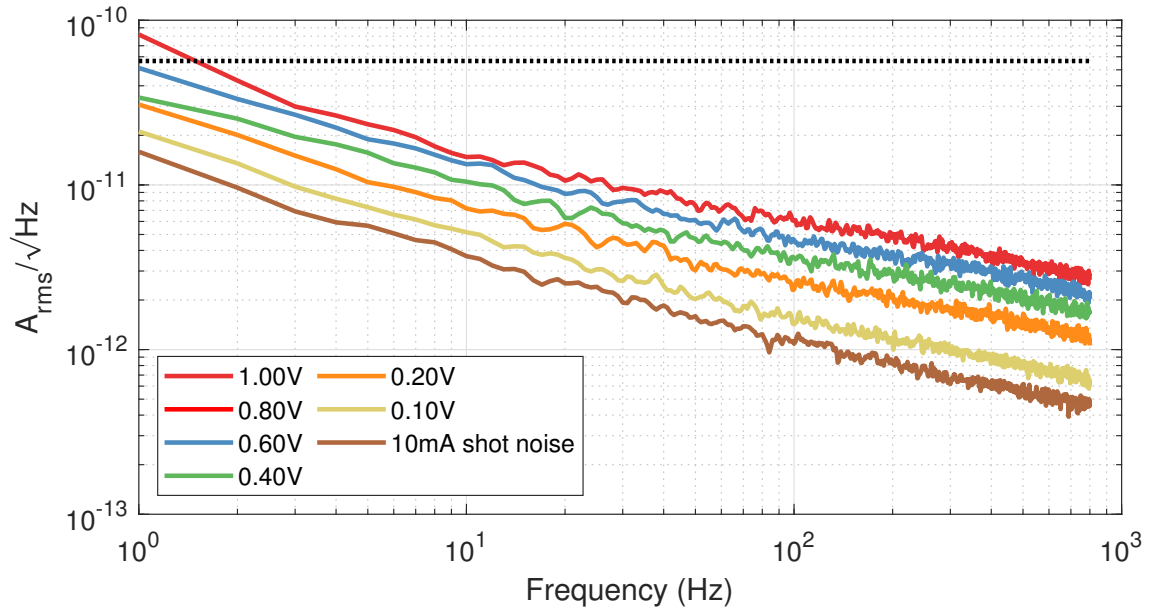


Figure A.6: G12182-010k

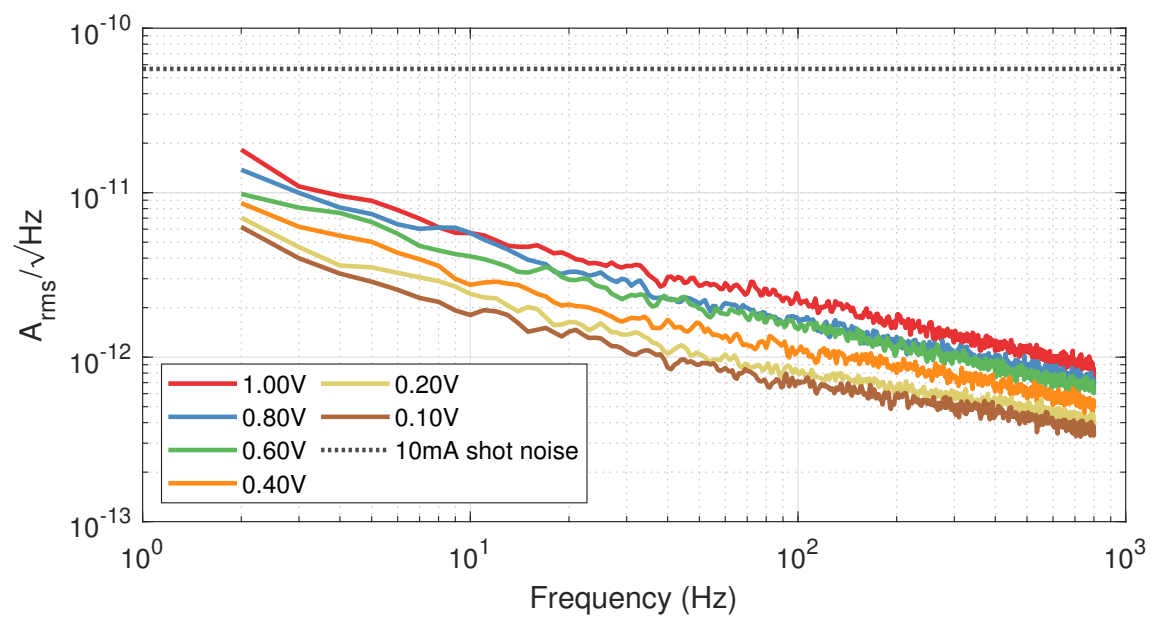


Figure A.7: G12182-005k

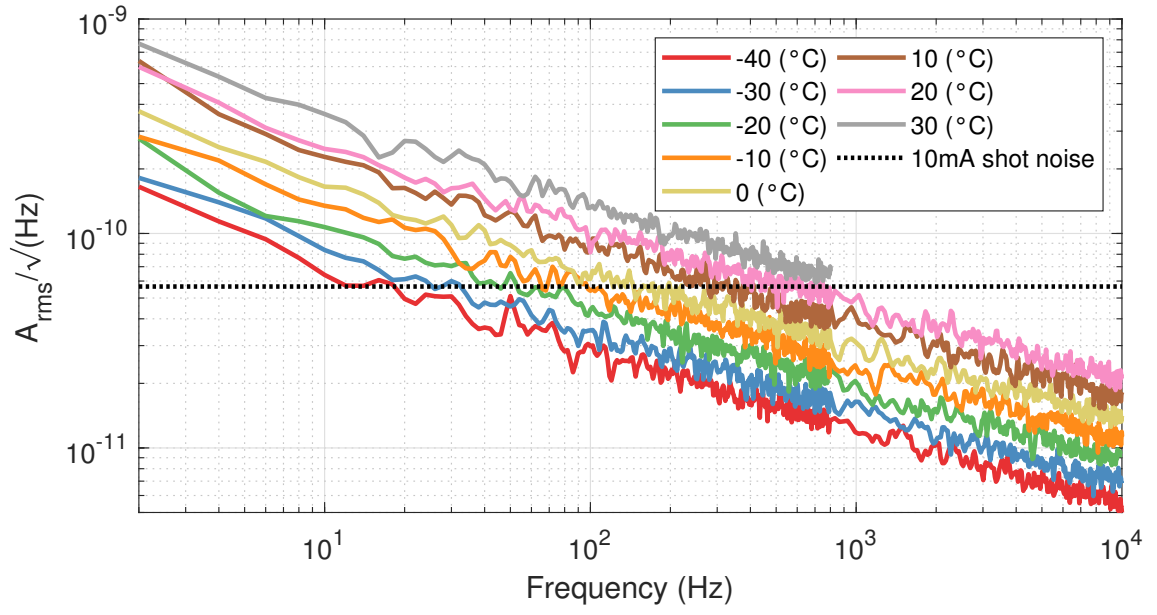


Figure A.8: FD10D 1.8V reverse bias

## A.2 Temperature Section

## A.3 InSb

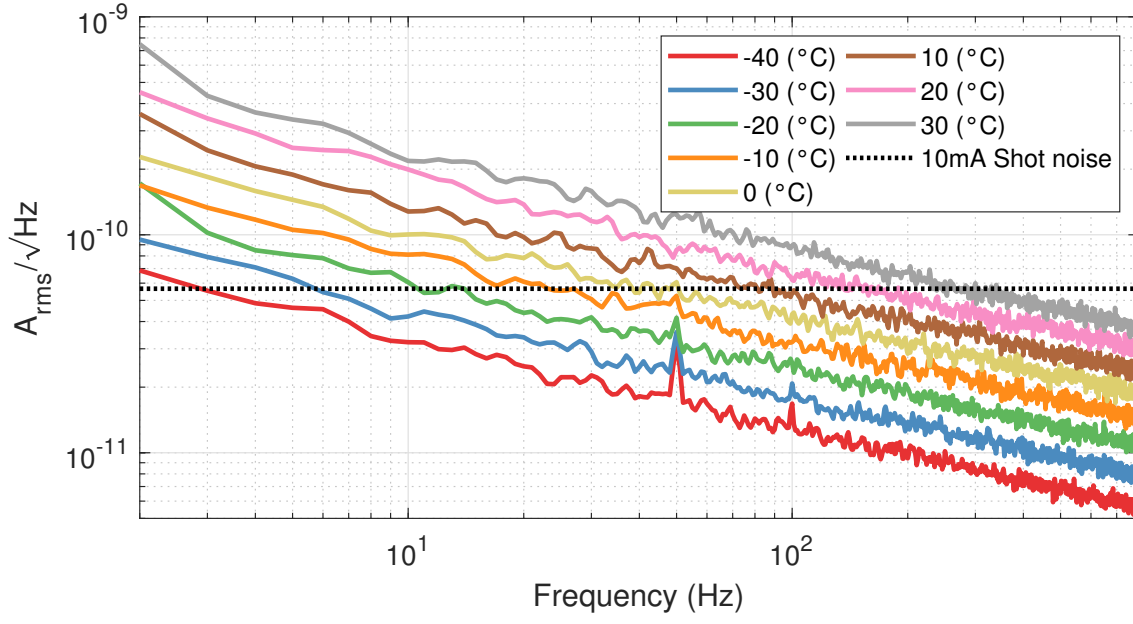


Figure A.9: G12183-010k 1.0V reverse bias

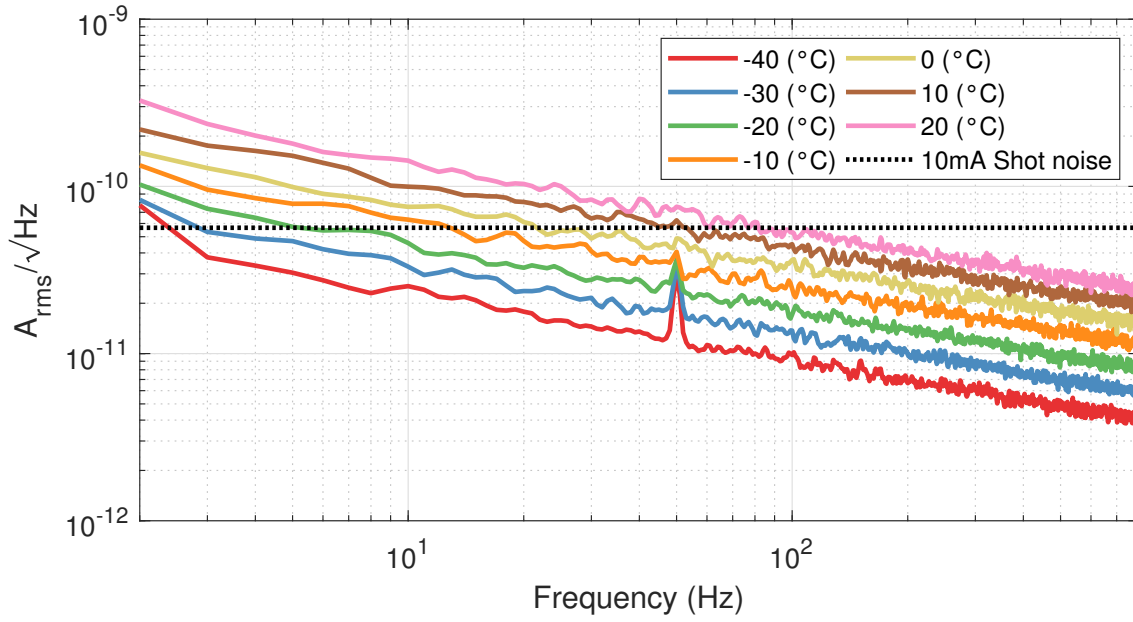


Figure A.10: G12183-005k 1.0V reverse bias

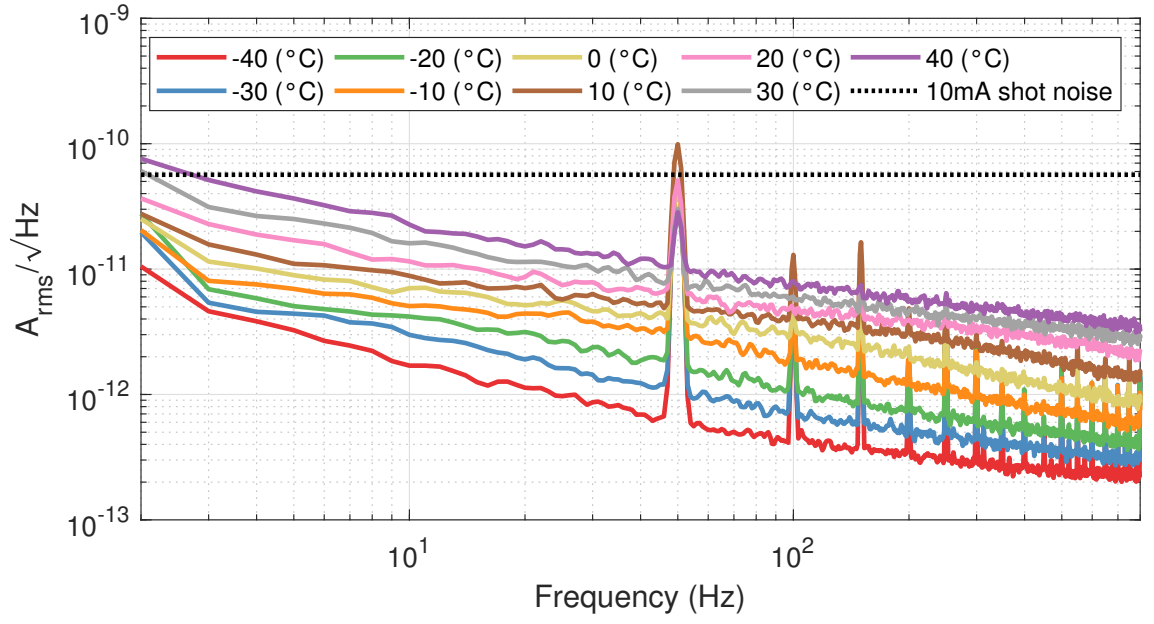


Figure A.11: G12182-010k 1.0V reverse bias

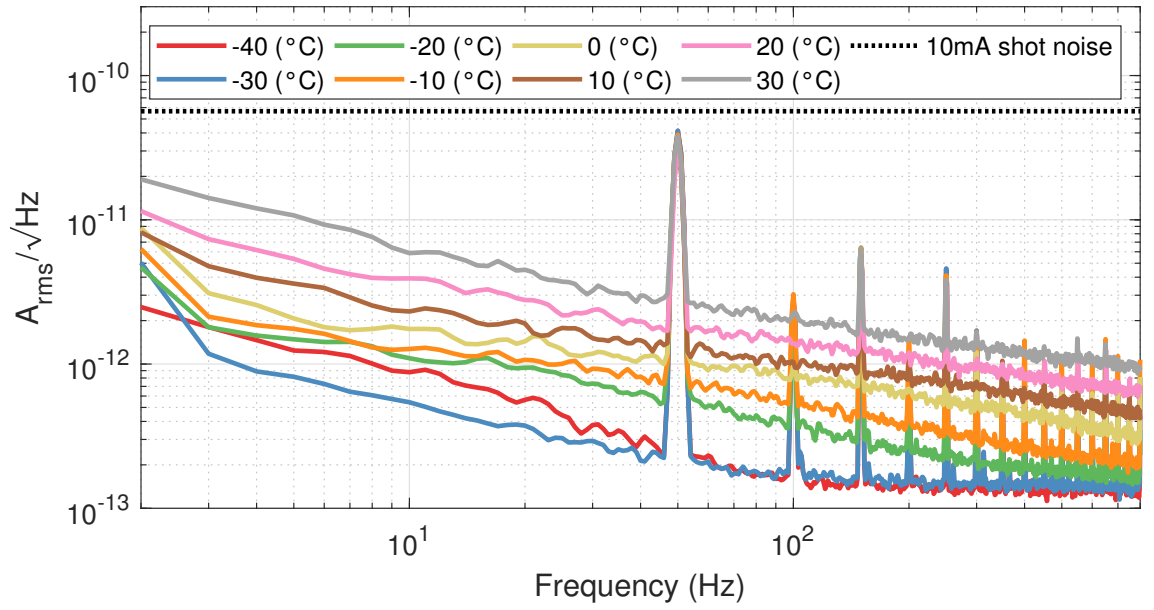


Figure A.12: G12182-005k 1.0V reverse bias

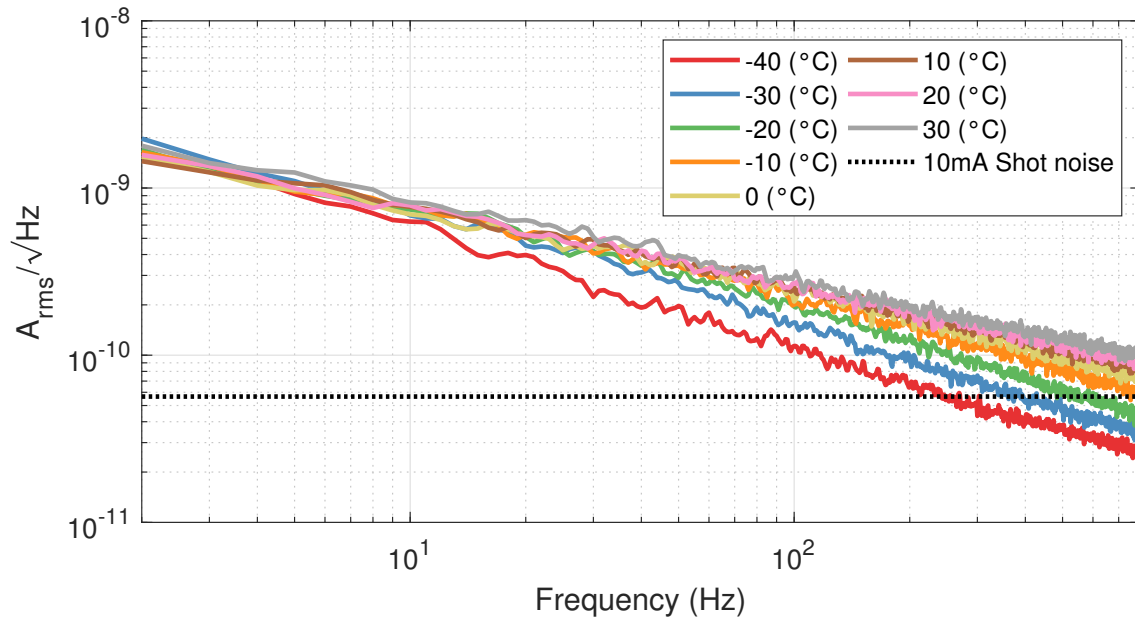


Figure A.13: IG24x500S4i 1.0V reverse bias

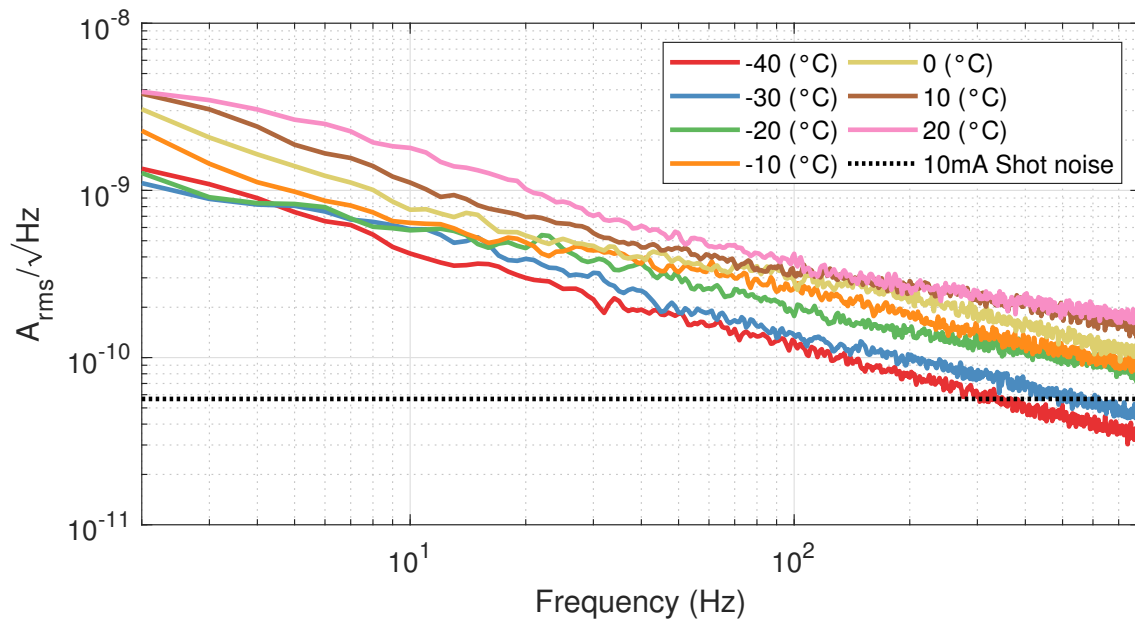


Figure A.14: IG26x500S4i 1.0V reverse bias

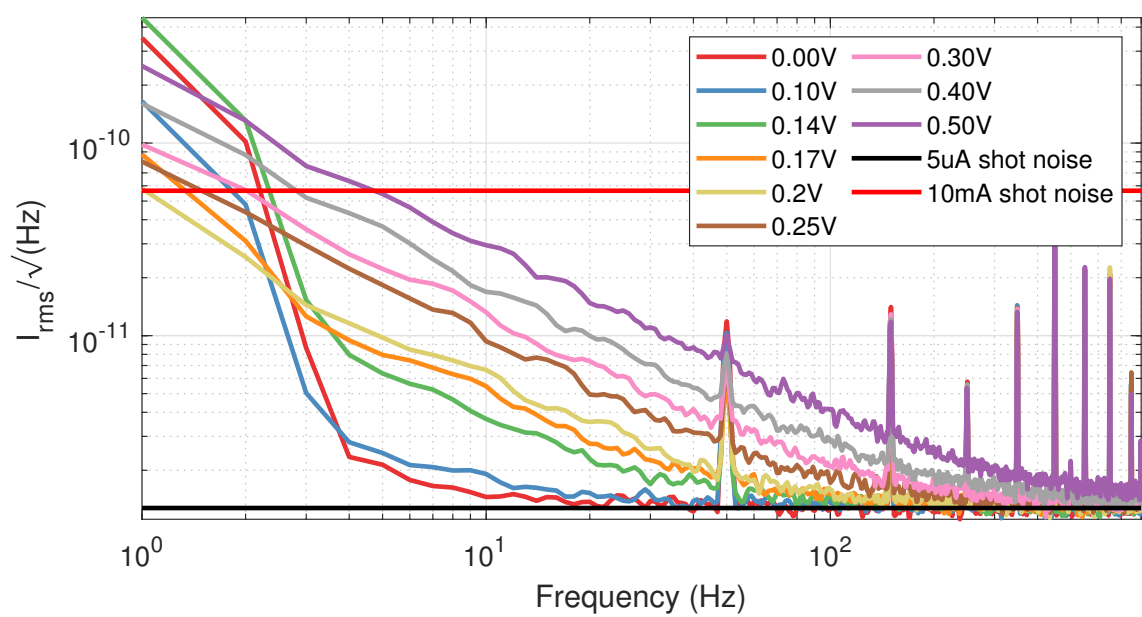


Figure A.15

# Appendix B

## Photodiode Circuits

### B.1 Transimpedance Amplifier for Measuring the Dark Noise of Photodiodes

A circuit which allowed for the reverse bias to be set and for the easy exchange of photodiodes was constructed. To measure small amounts of current noise, a transimpedance amplifier with a large gain is needed. As the dark current for these diodes is of the order of 1mA, the gain of the transimpedance amplifier was chosen to be  $1 \times 10^6$  so that signals of the order 1 V would be produced, and so a 1 M $\Omega$  resistor was used as in the feedback path.

Operation amplifiers have current noise at their inputs, voltage noise at the their outputs and the feedback resistor will generate Johnson-Nyquist noise (e.g. [27, Figure 8.58]). Designers of OP amps, generally, improve the OP amps output voltage noise at the expense of the input current noise or vice versa and so there is not one-size-fits-all op amp. In this application, as the current noise that is to be measured

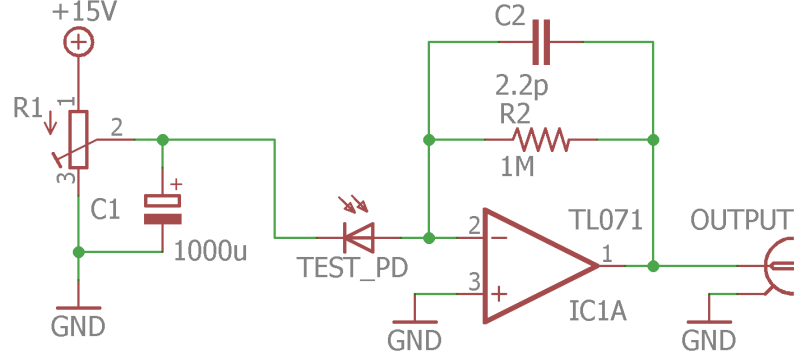


Figure B.1: Circuit used to measure the dark noise of photodiodes, represented by TEST\_PD, under different reverse biases. The transimpedance amplifier has a  $1\text{M}\Omega$  (R2) in parallel with a  $2.2\text{pF}$  capacitor (C2) in the feedback path of the TL071 op amp (TL071P). The reverse bias was generated by a potentiometer (R1) connected to an LM7815 voltage regulator. This voltage was low passed using C1

is small and the feedback resistor is large, a TL071, a JFET style op amp with low input current noise, was selected. As the photodiode has capacitance and the op amp has finite bandwidth, a  $2.2\text{pF}$  capacitor in parallel to keep the circuit stable.

Several methods of providing the reverse bias were tested. The first of these derived the bias from a LM7815 regulator in combination with a variable resistor configured a potential divider; this circuit is shown in Figure B.1. A similar circuit which was used in later experiments is shown in Figure B.2.

To verify that this was a sufficiently quiet bias voltage source that does not contaminate the measurements with noise, an ultra-quiet bias reference was used to cross check a measurement; this circuit is shown in Figure B.2. By filtering the voltage made a reference voltage IC (AD587JNZ), this circuit provided a bias with

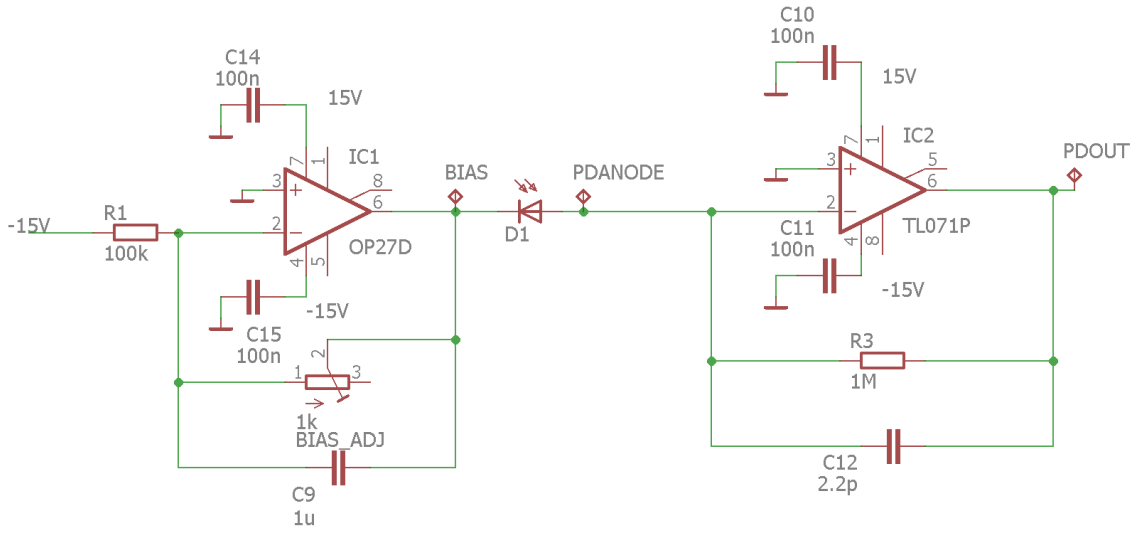


Figure B.2: Circuit used to measure the dark noise of photodiodes. The transimpedance amplifier used is identical to the one shown in Figure B.1. The bias voltage was generated using an OP27 op amp (OP27D) configured with so that the gain between the inverting input and the output of OP27D was  $> 1$ .

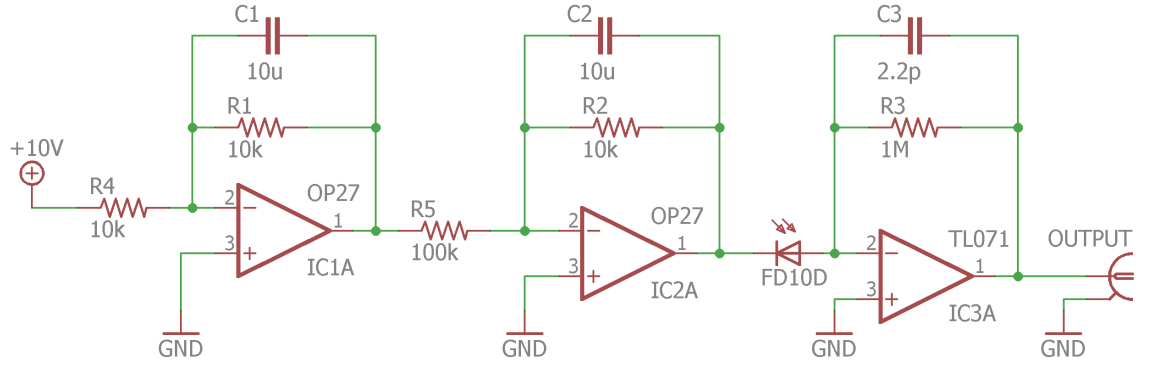


Figure B.3: Circuit used to check that bias voltage noise was not contaminating dark noise measurements. To reach  $5\text{nV}/\sqrt{\text{Hz}}$  at 10 Hz, a 10V low noise reference IC (AD587JNZ) was used to generate the reverse bias. The output of this IC was filtered by two low pass filters (IC1A and IC2A). The transimpedance amplifier used is identical to the one shown in Figure B.1

$5\text{nV}/\sqrt{\text{Hz}}$  at 10 Hz. The bias of the FD10D (see Section ??) was set to 1 V in both circuits. Figure B.4 shows that the regulator-potentiometer bias voltage source is sufficiently quiet.

### B.1.1 Transfer function

The transfer function of the transimpedance amplifiers shown in Figure ?? was measured under a variety of different physical constructions. This circuit's behaviour can be affected by stray capacitance. When constructed from strip board and with the photodiode anode and cathode cables very near each other, there was of the order 10 pF of stray capacitance i.e. it was as if the circuit had an extra 10 pF capacitor between the inverting input of the op amp and ground. By making this circuit on a prototyping board with less stray capacitance between the strips and

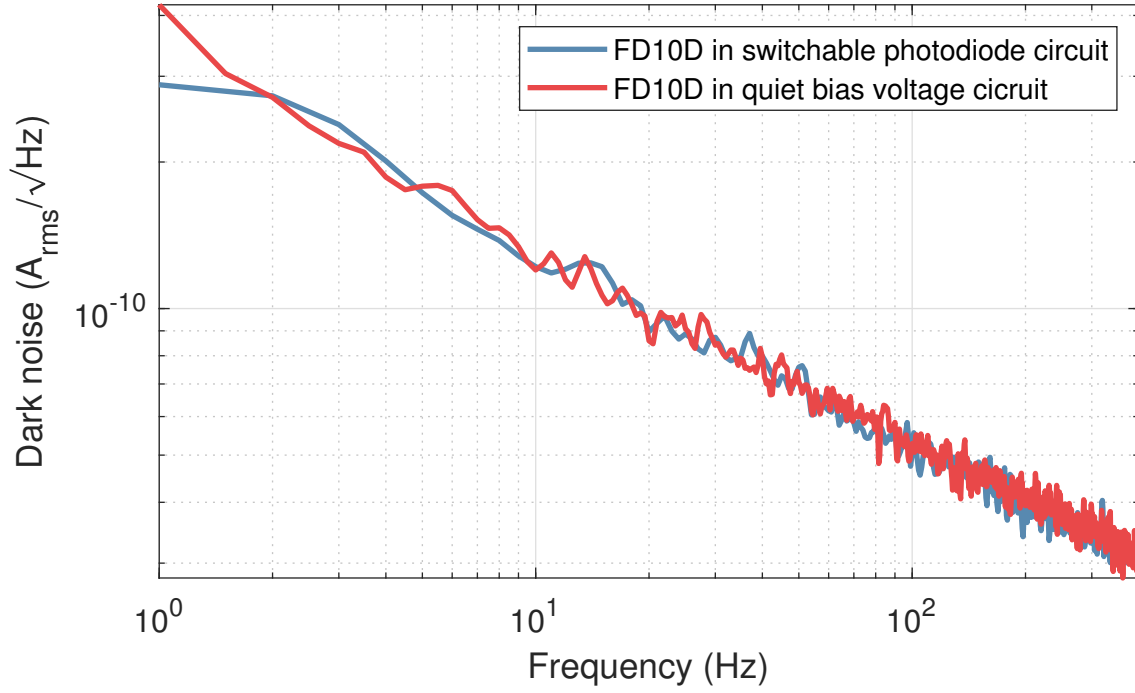


Figure B.4: The FD10D was biased with 1 V in the circuits shown in Figure B.1 and Figure B.2. The low noise fixed bias circuit and the circuit with a settable bias gave the same results, so we can be confident that the circuits shown in Figures B.1 and B.3 will produce measurements that are not contaminated by bias noise.

by keeping the photodiode anode and cathode wires from forming a capacitor, the stray capacitance was reduced. The measurement is in agreement with the LISO simulation of this circuit. This was cross checked using the method described in Section B.4.

### **B.1.2 Noise measurement**

The dark noise of the circuits described in Section B.1 is shown in Figure B.6. The shot noise level of a typical photocurrent in a gravitational wave detector is several hundred times larger than the noise of this circuit. To measure noise of the transimpedance amplifier, instead of a photodiode, a 470 pF capacitor was connected between the inverting input of the op amp and ground.

## **B.2 Transimpedance Amplifier for Making Shot Noise limited Measurements of 10 mA Photocurrents**

A photodiode circuit which is shot noise limited over a up to several 100 kHz is shown in Figure B.7. A DC bias was created by low-passed bench-top lab power supply. An AC component could be added onto the bias via a summing OP-amp. The FD10D was connected to a transimpedance amplifier that whose electronic noise was below the shot noise level for  $\sim 10$  mA currents. The noise of this circuit is shown in ... the transfer function of this circuit is shown in...

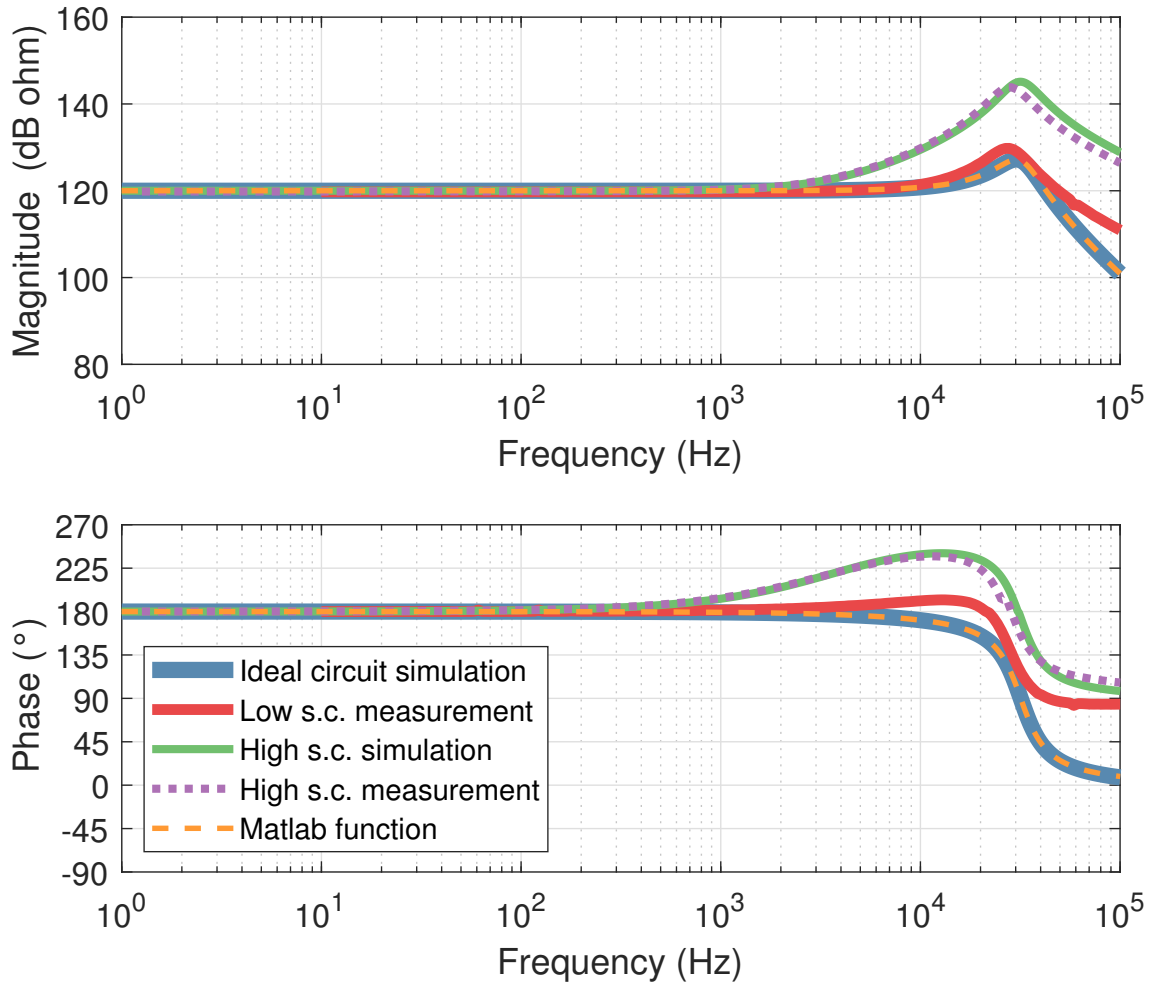


Figure B.5: The transfer function between a current into the inverting input of the transimpedance amplifiers designs shown in Figures B.1B.3 and B.2 was measured and simulated using LISO. A MATLAB function (see Section B.4) was used to check the measurement and the simulation. This circuit is sensitive to stray capacitance (s.c.) at the inverting input of the TL071. The stray capacitance was reduced by changing the layout of the circuit.

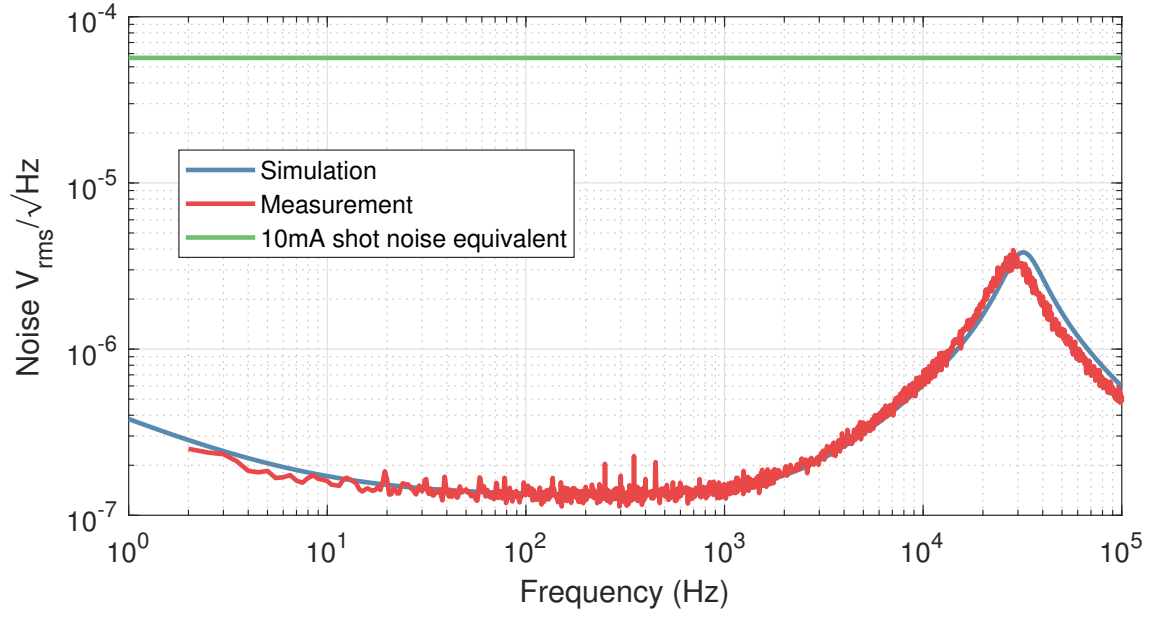


Figure B.6: At low frequencies, the noise of the circuit design shown in Figures B.1B.3 and B.2 is dominated by the low frequency voltage noise of the TL071. Above 10 Hz, the noise of the circuit is dominated by the johnson-Nyquist noise of the feedback resistor. At 30 kHz, the impedance of the feedback network causes the TL071's voltage noise to rise above the resistors noise.

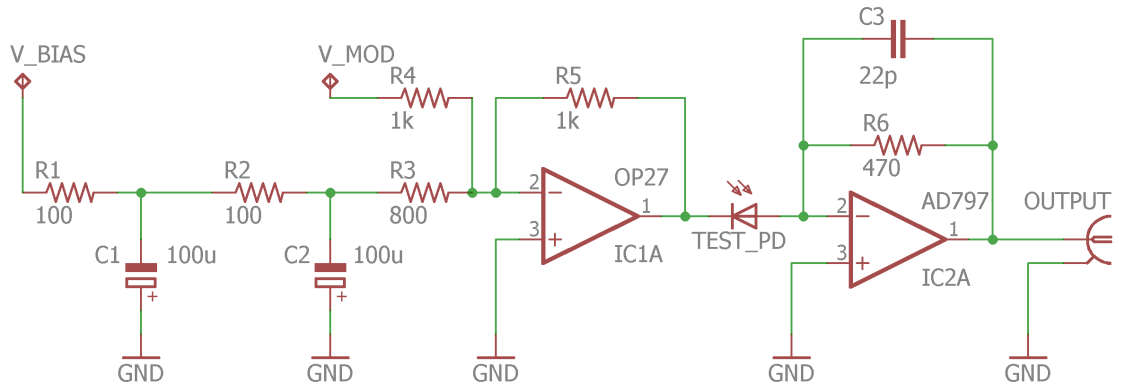


Figure B.7

## B.3 Simulations with LISO

The behaviour of circuits was simulated using LISO. This software performs a nodal analysis to calculate the voltages at each point within a circuit. LISO can calculate the noise and transfer function of circuits that include op amps. LISO has a wide variety of op amps in its op amp library and this can be easily expanded by specifying the noise properties, open loop gain and gain-bandwidth frequency.

## B.4 Transfer Function Calculation With MATLAB

The open loop transfer function of an op amp is

$$A(s) = \frac{A_0}{\frac{A_0}{2\pi f_{\text{GBW}}}s + 1}. \quad (\text{B.1})$$

The transfer function between the input current to the inverting input and the voltage at the output pin for an op amp made into a transimpedance amplifier is

$$H(s) = Z_{\text{FB}} \frac{A(s)}{1 + A(s)B(s)}, \quad (\text{B.2})$$

where  $B(s)$  is the transfer function of the components between the output pin of the op amp and the inverting input of the op amp and  $Z_{\text{FB}}$  is the impedance seen by the current flowing from the output pin to the inverting input. Without a capacitor in parallel with the feedback resistor  $B(s) = \frac{1}{R_{\text{FB}}C_{\text{PD}}s + 1}$ . This can be problematic as the transfer function of the circuit may have a phase of  $180^\circ$  due to the photodiode's capacitance and the gain bandwidth product of the op amp as it crosses the unity gain point which will cause the circuit to be unstable. To fix this problem, a capacitor

should be added in parallel to the feedback resistor, and so

$$B(s) = \frac{R_{\text{FB}}C_{\text{FB}}s + 1}{R_{\text{FB}}(C_{\text{FB}} + C_{\text{PD}})s + 1}. \quad (\text{B.3})$$

The impedance seen by the current is

$$Z_{\text{FB}} = \frac{R_{\text{FB}}}{R_{\text{FB}}(C_{\text{FB}} + C_{\text{PD}})s + 1}. \quad (\text{B.4})$$

$H(s)$  can be calculated with the following MATLAB code:

```
function [mag,phase] = photodiodeTransimpedance(cf, rf,cpd,gbw,A,w)
% cf is the capacitor in the feedback loop used to keep the circuit
% stable, and its units are Farads. rf is the feedback resistor
% measured in Ohms. cpd is the photodiodes capacitance (Farads).
% gbw is the
% gain bandwidth of the OP amp, and is measured in Hz. A is the
% Open loop
% gain of the OP amp. w is the angular frequency range that
% the phase and magnitude will be calculated over.
A_s = tf(A,[A/(2*pi*gbw),1]);
B_s = tf([cf*rf,1],[rf*(cf+cpd),1]);
closedLoopAnalytic = -tf(rf,[rf*(cf+cpd),1])*A_s/(1+A_s*B_s);
[mag,phase] = bode(closedLoopAnalytic,w);
mag = squeeze(mag);
phase = squeeze(phase);
end
```

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